The Arab American University

Faculty of Information Technology

Project Title: ResHub

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Date:

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Declaration

This is to declare that the graduation project entitled (ResHub) under the supervision of Dr. Mohammed Maree is our own work and does not contain any unacknowledged work or material previously published or written by another person, except where due reference is made in the text of the report.

Date: December 1, 2024

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Abstract

The research process often faces a variety of challenges, such as managing projects, literature reviews, and publishing findings. This proposal outlines the development of a fully integrated comprehensive, AI-powered platform designed to streamline and enhance the research lifecycle thoroughly, from idea conception to publication and beyond. Key features include project management tools for collaboration, AI-driven chatbot features like a chatbot for literature review discovery, critique tool, writing tools supported with grammar checks, and robust secure data storage, intelligent literature review search and annotation tools, and citation management. The platform also facilitates recommendation engine that connects researchers with other compatible researchers and propose compatible publishing topics, discussion forums for community participation, researcher profiles (with semantic graphs for highlighting interests) and generic journal finder. By distinguishing between guest, regular, and premium roles, ensuring versatile access to different layers of features, targeting researchers, academics, institutions and publishers as the primary audience, this platform will cater to all stakeholders involved in the research process, including students and independent scholars. What sets our platform apart is its unique combination of AI-driven insights, an intuitive user experience, ensuring efficiency. The platform will be developed using flask web framework technology, ensuring a fast, reliable, and universally accessible experience across devices. With security measures, accessibility features, and interoperability, this platform aims to revolutionize the way researchers work, enabling much more productivity, collaboration, and innovation.

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List of Abbreviations

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Chapter One. Introduction

1.1 Background and Motivation

Research involves many steps, like coming up with ideas, reviewing literature, writing papers, and getting them published. These tasks can be overwhelming and time-consuming, especially when researchers face challenges in organizing their work, sharing tasks with others, or keeping their data safe. Writing papers, checking grammar, finding journal, finding similar researcher, critique literature, managing citations, and going through the publishing process can also take a lot of effort.

With the help of artificial intelligence (AI), many of these challenges can be solved. AI tools can make research easier by providing literature review discovery and summary, finding gaps in critique literature review, ability to suggest best journal for you, quick annotations, grammar and citation help, and tools for working with different researchers through their semantic graphs. A platform that combines all these tools in one place can save researchers time and make their work more productive.

This project aims to build an AI-powered platform to help researchers manage their work from start to finish. It will make the research process faster, more organized, and less stressful. The goal is to create a tool that helps researchers focus more on their discoveries and less on the difficult tasks.

1.2 Aims and Objectives

Aim:

To create an AI-powered platform that makes the research process easier and faster, helping researchers manage their work, collaborate, and share knowledge.

Objectives:

Help with Writing and Publishing:
Include features like grammar checks, citation help, and easy options to submit research papers.

Make Information Easy to Find:
Use smart search and annotation tools to quickly find and organize data.

Encourage Teamwork:
Add networking tools, discussion spaces, and profiles to connect researchers and share ideas.

Keep Data Safe and Easy to Use:
Ensure strong security and simple, user-friendly designs for all researchers.

Offer continuous support and updates:

To keep users informed and share the latest developments in their disciplines, provide real-time notifications, and a dynamic user interface.

Promote a user-friendly experience:

Create a friendly user platform simple enough to used by people with different levels of technical knowledge, ensuring widespread acceptance and accessibility.

AI-enabled writing and critical tools:

Creating artificial intelligence (AI) systems that can assess the quality of writing research and make suggestions for improvements as well as checking grammar, helping researchers produce high-quality works.

Promoting cooperation among researchers:

To foster a community-oriented research environment, provide features such as researchers' profiles, semantic graphs and collaboration tools that allow researchers to communicate based on common interests.

Make writing research and publishing easier:

Give researchers access to AI-enabled tools to support writing, citation management and grammar correction and monitoring so they can produce and publish their work more easily.

1.3 Problem Statement

In this project, we focused on these research questions:

How can the proposed AI platform (ResHub) simplify the scientific research process from the beginning of the researcher's creation of the idea until reaching publication.

How can AI tools such as smart search, commenting tools, and checking grammar rules for multiple languages improve the efficiency and accuracy of researchers' work?

How to enhance the spirit of cooperation and participation among researchers through their use of communication tools and forums and working to publish their research on the General Journal Search Tool?

How the system reduces researchers' time when collecting and summarizing research papers related to their project?

How does the system protect user data and how secure is it?

How to verified performance, quality, expansion and increased usage?

Hypotheses:

This project is based on the following hypotheses:

This AI platform will greatly enhance the efficiency and productivity of the practical research cycle that researchers seek, which will reduce the time and effort of researchers and increase the accuracy and professionalism of the research they conduct.

Through smart search, commenting and writing support tools, the accuracy and quality of the research outputs that the researcher wants will be improved, thus increasing the professionalism of practical research.

When researchers use the forums feature and the general search feature for journals closest to the research, in addition to the most expensive use of the semantic graph feature that works to find users who are similar to the researcher, innovation and knowledge sharing will be enhanced because a more connected research environment will be created.

The system will be built to be easy to use so that any individual from different communities will be able to use the system and adopt it in research.

Users' private data and profile information will be fully protected through platform data encryption, user authentication and compliance with data protection rules.

1.4 Contributions

There are many challenges that face the researcher during the research process, the most important of which are project management, finding relevant research, comparing it, finding strengths and weaknesses, and publishing the results. Therefore, the ResHub project aims to:

- It contains tools supported by artificial intelligence: The project provides advanced features such as intelligent search in literary references, citation management, and works to assist writing by checking grammar and writing style in addition to the semantic graph feature that allows the user to find other users who are related to his features and the general magazine searcher that works to find the best magazine for the user to publish his research in.
- Enhancing collaboration and communication among researchers: ResHub allows researchers to create a profile and provides a forum feature that allows researchers to participate in, which enhances the spirit of collaboration among researchers not only locally but also globally.
- User-friendly design with easy accessibility: The project will use Flask technology, an advanced web application that will make the ResHub experience easy, reliable, and compatible with all different devices.
- It is used in several stakeholders: ResHub is not only targeting researchers but also academic institutions, publishing houses, and funding agencies, enabling collaboration between different stakeholders in one integrated platform.
- Transforming Search: ResHub improves traditional search methods by combining intuitive traditional writing with AI technologies, which in turn increases productivity and creativity.

1.5 Structure of the Report

This report is organized as follows:

Chapter One: Providing an overview of the project and what are the motivations, aims and objectives for this project, in addition it will provide the main subject of the problem, raising Common questions and answers, as well as highlighting the contributions that the project serves, what are the key features, and the what makes it appealing to users.

Chapter Two: Providing an overview of existing projects similar to this one, highlighting their strengths and weaknesses.

Chapter Three: Defines the functional and non-functional requirements for developing an AI-powered solution.

Chapter Four: Presenting the system's structural design, including class diagrams, database schemas, and user interfaces, focusing on usability and efficiency to meet defined requirements.

Chapter Five: Highlighting the project's achievements in addressing research challenges, outlines unresolved issues, and proposes future enhancements to expand functionality and user engagement.

Chapter Two. Literature Review

2.1 Overview

ResHub is a platform using AI to improve research cycle, it introduces tools to help researchers to elevate their research outcomes, tools such intelligent literature review search, annotation tools, and citation management. They can also communicate with other researchers through forums and recommendation based on personal profiles, a researcher can create an account and write their research with the help of writing and critique tools till publishing their research, it targets researchers, academics and students and works to increase the spirit of cooperation and improve results, through a friendly user interface.

In this second chapter, we mentioned some systems that are similar to the ResHub, and we gave a brief explanation of each of the systems' features, how they work, their purpose, and their comparison with our project. We also mentioned the added features of the project that make it distinct from the similar systems that were mentioned.

2.2 Existing Systems

Many systems exist that provide similar functionalities to our proposed platform, "ResHub." These systems serve as benchmarks for developing a more integrated and AI-driven solution. In the following discussions, we will compare these existing platforms to highlight what differentiates "ResHub" from them, focusing on our unique approach to leveraging artificial intelligence to streamline project management, collaboration, and publication processes for researchers and academics.

1. mendeley.com:

Mendeley is a global website focused on reference management helping researchers organize their research, offering an academic social network that allows collaboration with others online, and discovering the most recent papers. The primary targeted audience are researches academics and students. The platform provides a variety of services like Reference Management, Citation Tool, Collaborative Tools, Research Network Features, Research Discovery Tool and Cross-platform availability. Despite these features the website lacks AI integration into the system limiting its capabilities and Inefficient Literature Management that can be efficiently improved by the existing of AI as a feature, and the absence of specific needs adaptation and preferences of individual users.

A picture of the site's main page will be placed below to see the site's design.

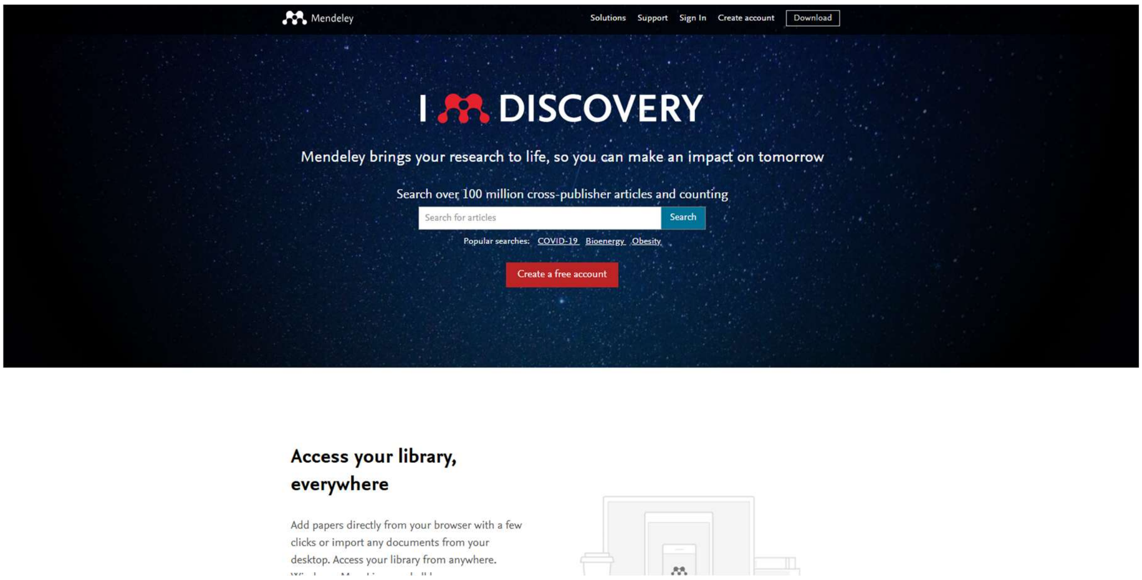


Figure 2.1 "A screenshot for the Homepage of Mendeley website"

2. ResearchGate:

ResearchGate is network for researchers and scientists, the scientist, clinician, student, engineer, public health worker, lab technician, computer scientisthe developers started in 2008 to solve the problems they faced in creating and sharing science, they connected the world of science and made research public to everyone, allow Researchers to create profiles to show their work, allows uploading and sharing of research articles, preprints, and datasets, promoting open access, provide Connect the researchers together and Join discussions, can the user give feedback, provides stats on how often your research is viewed, downloaded, or cited, Provides job opportunities in a special jobs section.

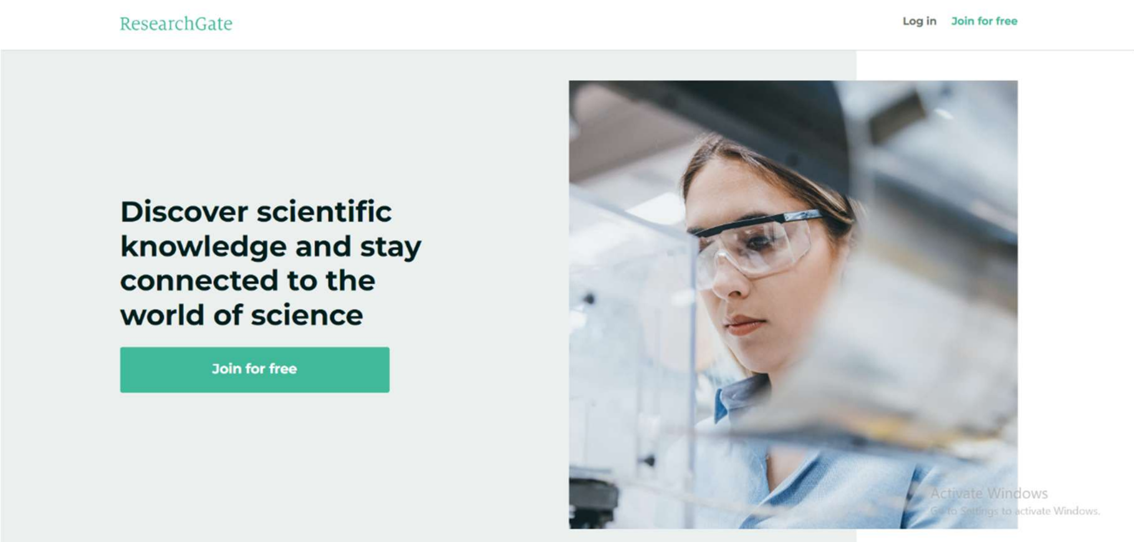


Figure 2.2 “A screenshot for the Homepage of ResearchGate website”

3. Semantic Scholar:

The system provides free search and discovery methods based on artificial intelligence, and provides users with open resources for the global research community, It has artificial intelligence search that makes the search more intelligent and accurate, It summarizes research papers to make them easier for the user to understand, Highlights the most significant citations within a paper to showcase its academic impact, research papers are classified by topic to facilitate navigation within the research, provides the researcher with research paper suggestions related to the research he is familiar with, It targets academic researchers, students and professionals across various disciplines.

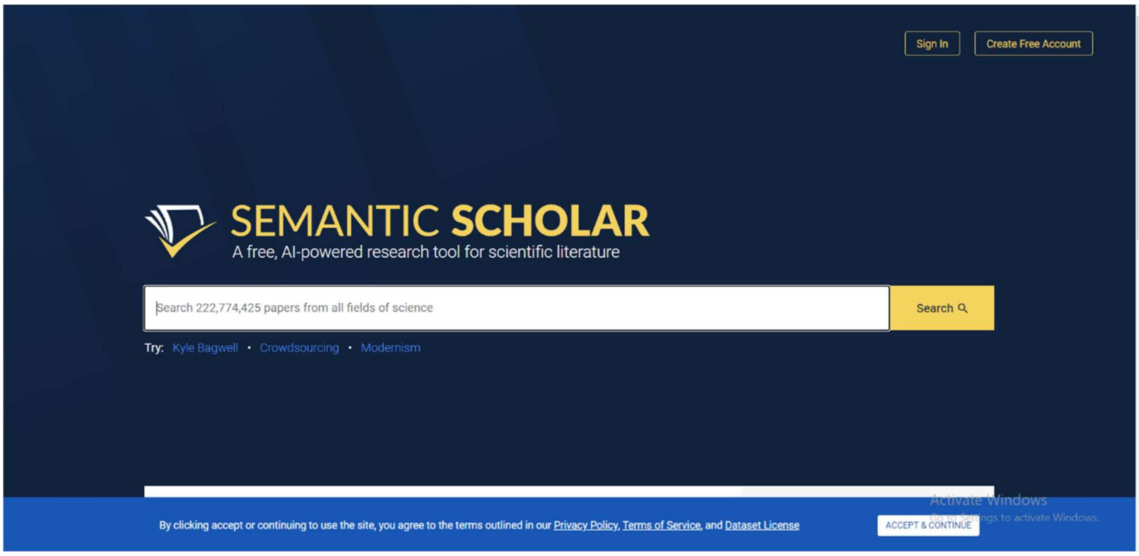


Figure 2.3 “A screenshot for the Homepage Semantic Scholar website”

4. Overleaf:

Overleaf a collaborative cloud-based LaTeX editor used for writing, editing and publishing scientific documents. It was founded in 2012 by John Lees-Miller and John Hammersley and has since become a popular tool for researchers, students, and professionals. Overleaf provides real-time collaboration, version control, and pre-built templates for various journals and conferences. It partners with a wide range of scientific publishers to provide official journal LaTeX templates, and direct submission links. The platform simplifies the use of LaTeX, a typesetting system commonly used for technical and scientific writing, by offering an intuitive interface and live previews of the output. Overleaf integrates with reference managers like Mendeley and Zotero and allows direct submission to many academic publishers.

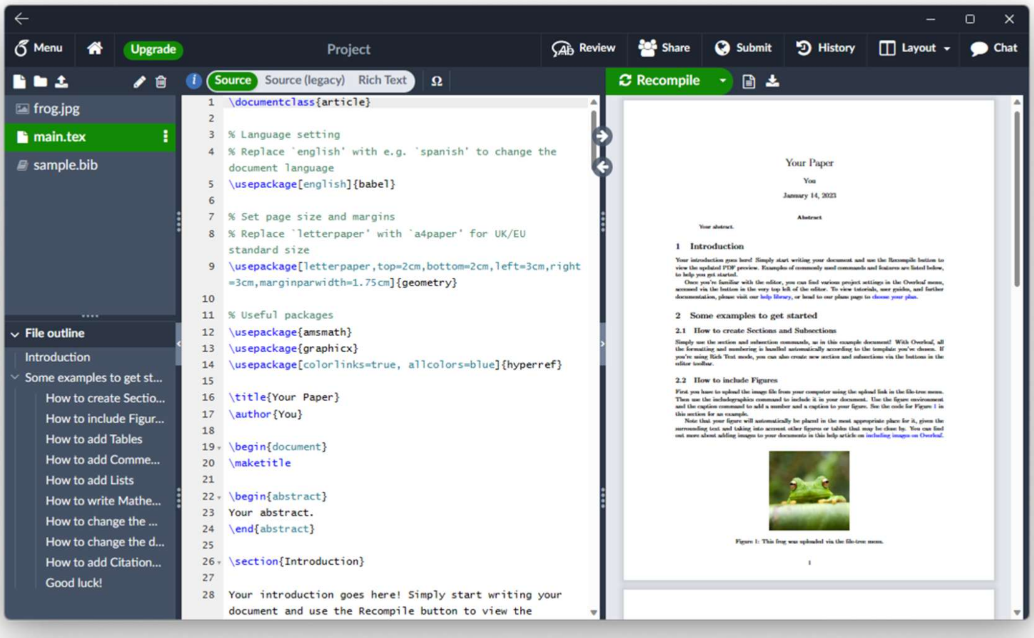


Figure 2.4 “A screenshot for the Homepage of Overleaf website”

5. Iris.ai:

Iris.ai is an AI-powered platform designed to help researchers with literature discovery, content analysis, and research exploration. It uses Natural Language Processing (NLP) and machine learning to understand scientific texts, identify relevant research papers, and provide insights from vast academic databases. The platform’s AI can analyze academic papers, extract key information, and help researchers navigate the growing body of research literature more efficiently.

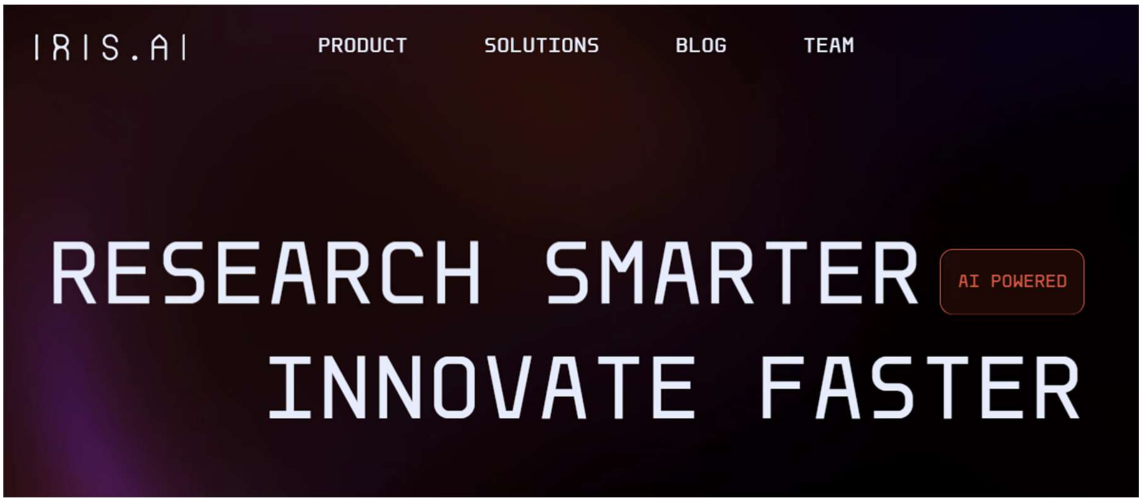


Figure 2.5 “A screenshot for the Homepage of Iris.ai website”

6. Google Scholar:

Google Scholar is a thorough search engine that comprehensively serves researchers, academics and students by offering a simple tool to search for literature across different disciplines and sources. The platform lists articles, theses, books and other literature written by researchers. Google Scholar offers numerous key features, including search functionalities spanning over various fields of study, citation tools, personalized alerts based on research interests and scholar metrics. Despite these the platform does not offer collaboration tools between users, nor incorporate AI-driven features enhancing personalization or literature management.

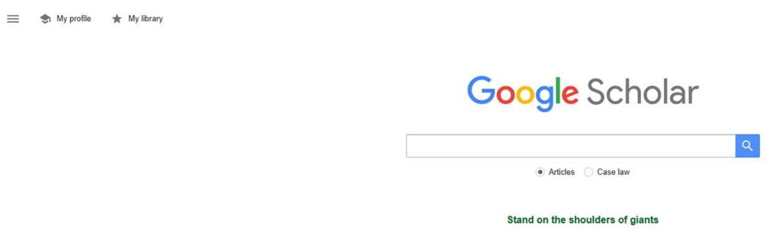


Figure 2.6 “A screenshot for the Homepage of Google Scholar website”

2.3 Comparison:

"Below is a table comparing our system with existing systems, mapping equivalent features and the information each system delivers. We've indicated strengths, weaknesses, and emphasized what distinguish our platform apart from other systems."

Site/comparison factors	ResHub	Research Gate	Semantic Scholar	Mendeley	Google Scholar	Iris.ai	Overleaf
Website/Application	website	Website Has also mobile application	Website Has also mobile application	Website Has also mobile application	Website	Website	Website
Language Language	English	English	English	English	Multilingual	English	Multilingual, but mostly used in English
Target audience	Researchers, Academics	Scientists, Clinicians, Students, Engineers, Public Health Workers, Lab Technicians, Computer Scientists.	Researchers and Scientists, Students, Academics and Professors, Industry Professionals, Policy Makers and Analysts, Librarians and Research Administrators.	Researchers, Students, Academics	Broad academic audience	Researchers, Academic Institutions, Scientific Professionals.	Researchers, Students, Academics (especially in technical fields).
Registration fees	Free tier available, paid plans for premium features	Free	Free	Free	Free	Free trial, paid subscription required for full access	Free tier available, paid plans for additional features
AI Integration	Advanced AI	None	None (Advanced search algorithms)	None	None	Strong AI capabilities for literature review	None
Availability of Chatbot	Available	Don't have chatbot	Don't have chatbot	Don't have chatbot	Don't have chatbot	Don't have chatbot	Don't have chatbot
Liturture Review	AI powered Search	The feature exists	The feature exist	The feature exist	The feature exist	The feature exists for literature discovery	Basic Search
site sections	Profile pages, research papers, Forums, liturture review discovary, critque, recoomendation page, journal finder	Filters for authors, journals, topics	Library management, reading list, groups	Filter by articles, authors, cited by	profile, library, alerts, metrics,advanced searchj	Document editing, templates, collaboration tools	Text Editing and Formatting, Publication Management, Collaborative Workspace, Discussion Rooms, Literature Review Search, Profile.
Collaboration and Community	Social networking, group discussions, collabrotive workspace	The Feature exist	The Feature exist	Shared libraries, collaborative annotation	None	Limited collaboration (mainly for institutions)	Excellent collaboration features with real-time editing and community access
Semantic Graph Profiles	Yes	None	None	None	None	None	None
Generic Journal Finder	Yes	None	None	None	None	None	None
Research Metrics & Document Stats	Yes	Yes	Yes	None	Yes	None	None
Publication	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 2.1: Comparison between studied Systems.

2.4 Summary

Chapter Two of our project report focuses on reviewing similar systems to our proposed "ResHub." It details various platforms that share functionalities with "ResHub," highlighting their strengths and weaknesses through a comparative analysis. This comparison helps us refine "ResHub" into a more integrated and AI-enhanced system. We discuss how each existing system functions, their main purposes, and their user interfaces, emphasizing how "ResHub" uses artificial intelligence to improve project management, collaboration, and the publication process in academic research. The goal is to show how "ResHub" stands out by more effectively enhancing the research lifecycle with advanced AI features.

Chapter Three: System Analysis

3.1 Introduction:

In this chapter, we introduce the system architecture and analyze the requirements (administrator, registered user, premium user, guest) and registration, all of which are essential steps to build the system. We provide the data information to the website to be build such as the sequence diagram, use case description, and the use of semantic graph to link the user to another researcher based on semantics, in addition to the possibility of the premium user to access the best suitable journal for his research and publish it there through the general journal search tool feature.

3.2 Analysis of Existing Systems

when we analyzed the requirement modeling, we searched for the most similar systems to the system we will build. We got to know it and how to use it, what it offers and what its weaknesses and strengths are, through which the requirements of our system became clearer. We analyzed the information that was collected and then divided it into functional and non-functional requirements as mentioned in the next section of system requirements.

3.3 System Requirements

This section shows system requirements and their explanations will be provided, highlighting their importance. These are divided into: functional requirements and non-functional requirements.

3.3.1 Functional Requirements

Functional requirements define what a system should do, describing the main features and abilities it must have to meet user needs and accomplish its objectives.

Admin:

- The Administrator can login
- The Administrator can access admin-specific pages
- The Administrator can view list of all user accounts with different types (Guests, Regular, Premium).
- The Administrator can suspend or block all users violating site rules.
- The Administrator can add, edit, or remove content on the site including enabling/disabling sections of the site
- The Administrator can control feature access for all type of users
- The Administrator can delete offensive discussion threads, remove improper posts or comments and pin important announcements.
- The Administrator can alter website theme elements such as colors, logos and pictures, header/footer text.
- The Administrator can push emergency notifications or announcements such as new feature release, planned downtime.
- The Administrator can view system logs

Guest:

- The guest can only read research papers.
- The guest can watch tutorial video about the system.
- The user can register to create an account by filling: username, password, and email address.
- The guest will be required to sign-up for any additional usages.

Regular Registered User:

- The user has the ability to create a personal research profile that is identified through a semantic graph containing their info
- The user can view and update personal information in their own account.
- When a user logs in, the system will view to them a home page.

- The user will be able to visit other researcher’s personal profiles (other users) and search with their names to find them (search functionality).
- The user can lookup how many times their document is read, downloaded, commented on, or cited and other users can do when opening the paper.
- The user can receive Mentions and Multi-channel notifications Email, in-website.
- The user can participate in forums discussions.
- The user will have the tools to write their research paper
- The user has the ability to publish their research
- The user can access the chatbot that can find and summarizes literature review in a professional way using AI capabilities.
- The user can collect related works (literature review) with citation file ready to use.
- The user can use AI tool to critique literature review and find gaps in them
- The user can use AI professional grammar checker while writing their paper.
- The user will be able to send feedback after using the system.
- The user will be shown additional features and be requested to sign-up for premium plan for any advanced additional usages.

Premium Registered user:

- The premium user can use a tool that meets them with recommended researchers according to their semantic graph.
- The premium user has the ability to access collaborative features with chosen recommended researchers.
- The premium user will be recommended with best magazine for their research (generic journal finder)

3.3.2 Non-Functional Requirements

Non-functional requirements determined the qualities and characteristics of the system, for instance performance, security, and scalability, which are essential for its overall success but don't pertain to specific features or functionalities.

Usability:

Users with different technical backgrounds should be able to easily navigate and get information from the platform as a result of its straightforward and user-friendly design.

Accessibility:

Obvious and simple layout to help users to move easily in the platform and find any information they want. Use accessible data visualization techniques such as graphs, to ensure all users can understand the represented data. Maintain consistent navigation mechanisms throughout the platform to help users predict where they can find information and how to move between sections.

Reliability:

The system needs to function dependably, reducing the likelihood of mistakes or malfunctions that can interfere with user interactions.

Performance Efficiency:

The system ought to be able to handle a huge number of concurrent users without any weakness in performance. Response time for user interactions must be within acceptable limits to ensure a smooth user experience. Optimize database queries and indexing to ensure quick data retrieval and minimize latency. Regularly analyze and refine database performance.

Data Integrity:

Reduce discrepancies and guarantee the accuracy and timeliness of available data provided to consumers.

Scalability and Flexibility:

We will design the system in a standard way so that if there is expansion in the future or the number of users is somewhat large or we add features, this will not affect the system's performance or disrupt it.

Compliance and Security:

The system design includes the General Data Protection Regulation that protects user data and also includes the Payment Card Industry Data Security Standard that protects user payment data. The system must provide protection for data from hacking by encrypting data in transit or at rest, controlling access according to priority, using strong authentication mechanisms such as (MFA)و and working on backup to ensure data integrity.

Performance Constraints

Response Time Constraint:

The system must respond to user queries via the chatbot within a maximum of 30-120 seconds to ensure a seamless user experience.

Resource Limitations:

ResHub contains AI tools that analyze literature reviews, check grammar, and generate citations. The system will use high CPU and run AI tools in parallel, which is very important to reduce slowdown.AI requires large datasets and linking them to large memory and using memory strategies to avoid system crashes. The storage capacity must be able to accommodate the research content created by the researcher and expand it as the user uses the system more. There are some users who have weak internet frequency (limited internet connection) and the system works to deal with this problem through such as downloading simplified user interfaces or reducing data loads for these users.

Scalability:

The system must handle the largest possible number of users at the same time without any slowdown or system failure.

Uptime and Availability:

The system must achieve a minimum uptime of 99%, ensuring that the system is always available to users with minimal system downtime during maintenance.

Security Measures:

Implementing strong security protocols through which user data is protected, and ensuring conformity with data protection laws and standards.

3.4 Design Objectives/Requirements (Optional)

3.4.1 System Architecture

In our proposed system, we employ a Flask-based web architecture, where user requests (from both Regular and Premium roles) flow into the application via defined routes rather than a strict MVC pattern. Each request triggers Flask route handlers, which process any required logic—such as user authentication, AI interactions, or data retrieval—and then return a response. The View portion is rendered through Flask templates (e.g., for dashboards, collaborative workspaces, or chat interfaces), while the Model layer covers database operations, research data, and chatbot logic. External services or security measures (like integrating with AI APIs or encryption modules) are handled within these route handlers or dedicated utility classes, ensuring that the system remains modular and can scale as more advanced features (like AI-driven recommendations) are introduced.

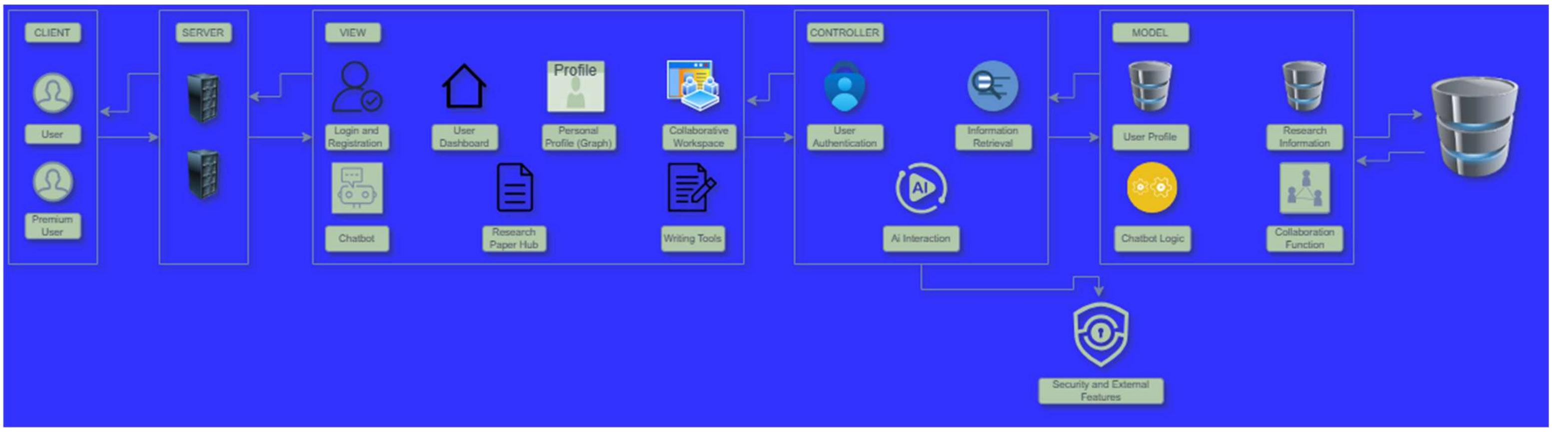


Figure 3.1 System Architecture Diagram

3.5 Development Methodology

3.5.1 Use Case Diagrams

A use case diagram represents the relationship between users and the system, illustrating how the system responds to various scenarios. It provides a simplified overview of system functionalities and helps identify user interactions.

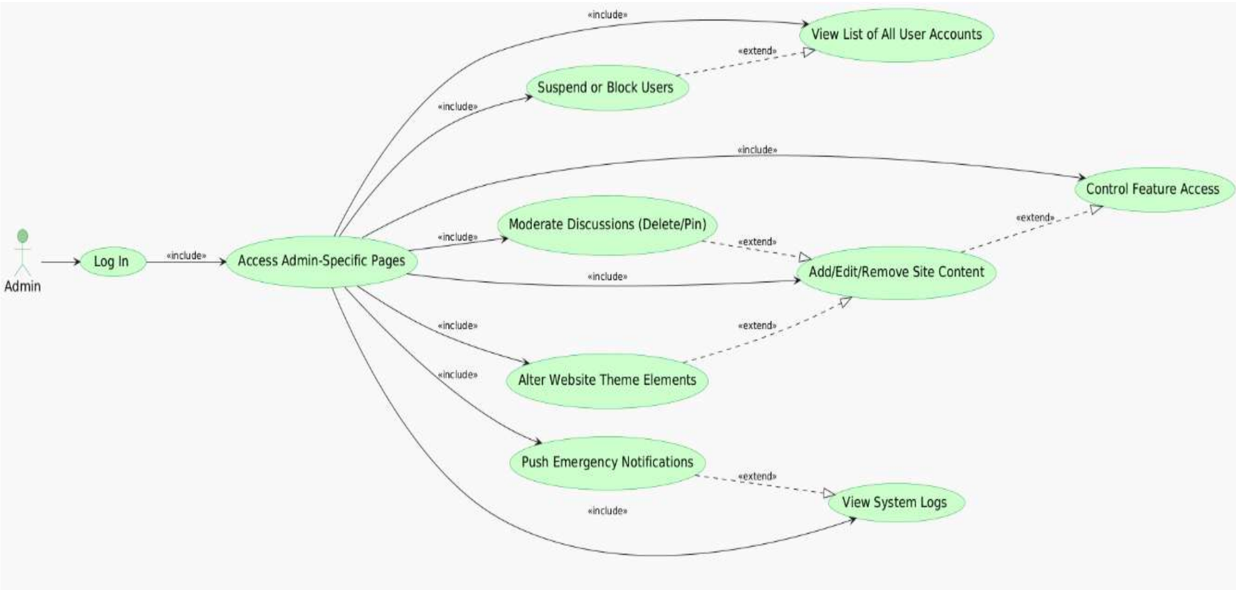


Figure 3.2 Admin use case diagram

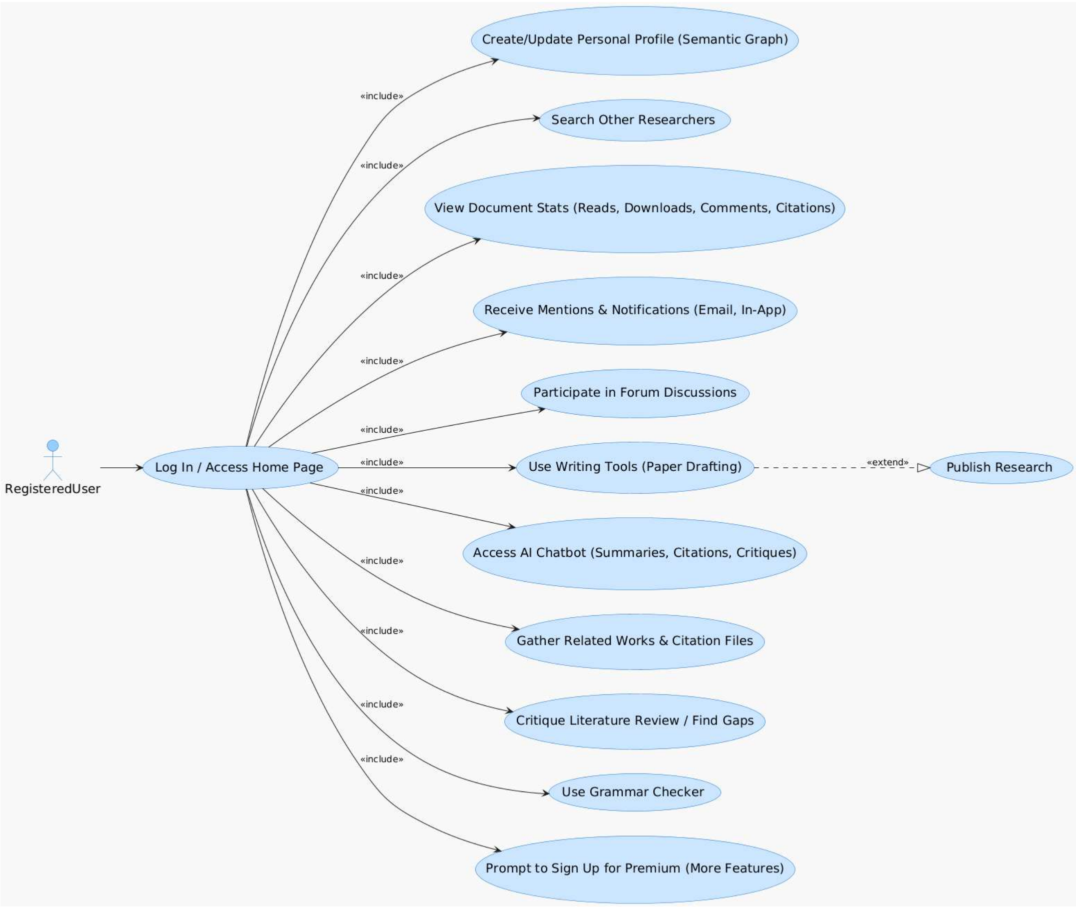


Figure 3.3 Registered user use case diagram

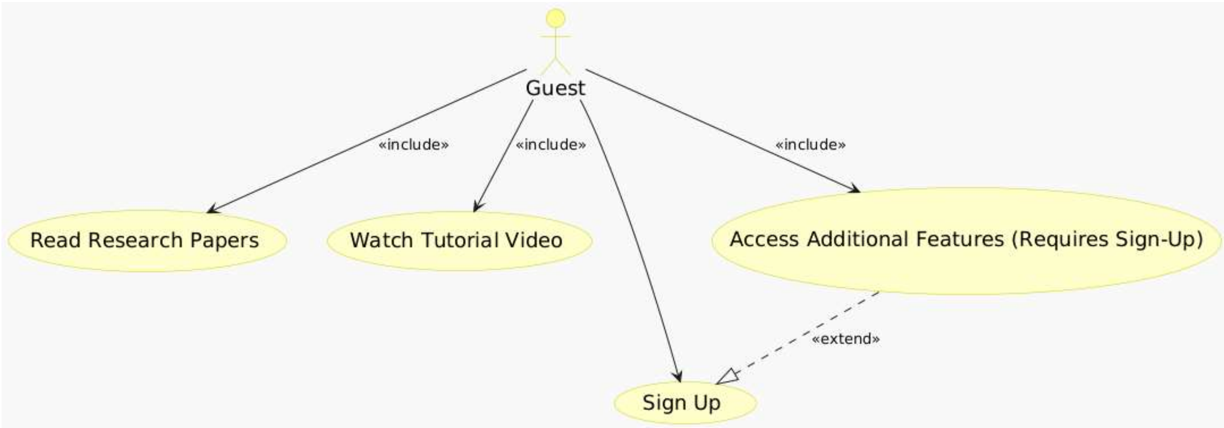


Figure 3.4 Guest use case diagram

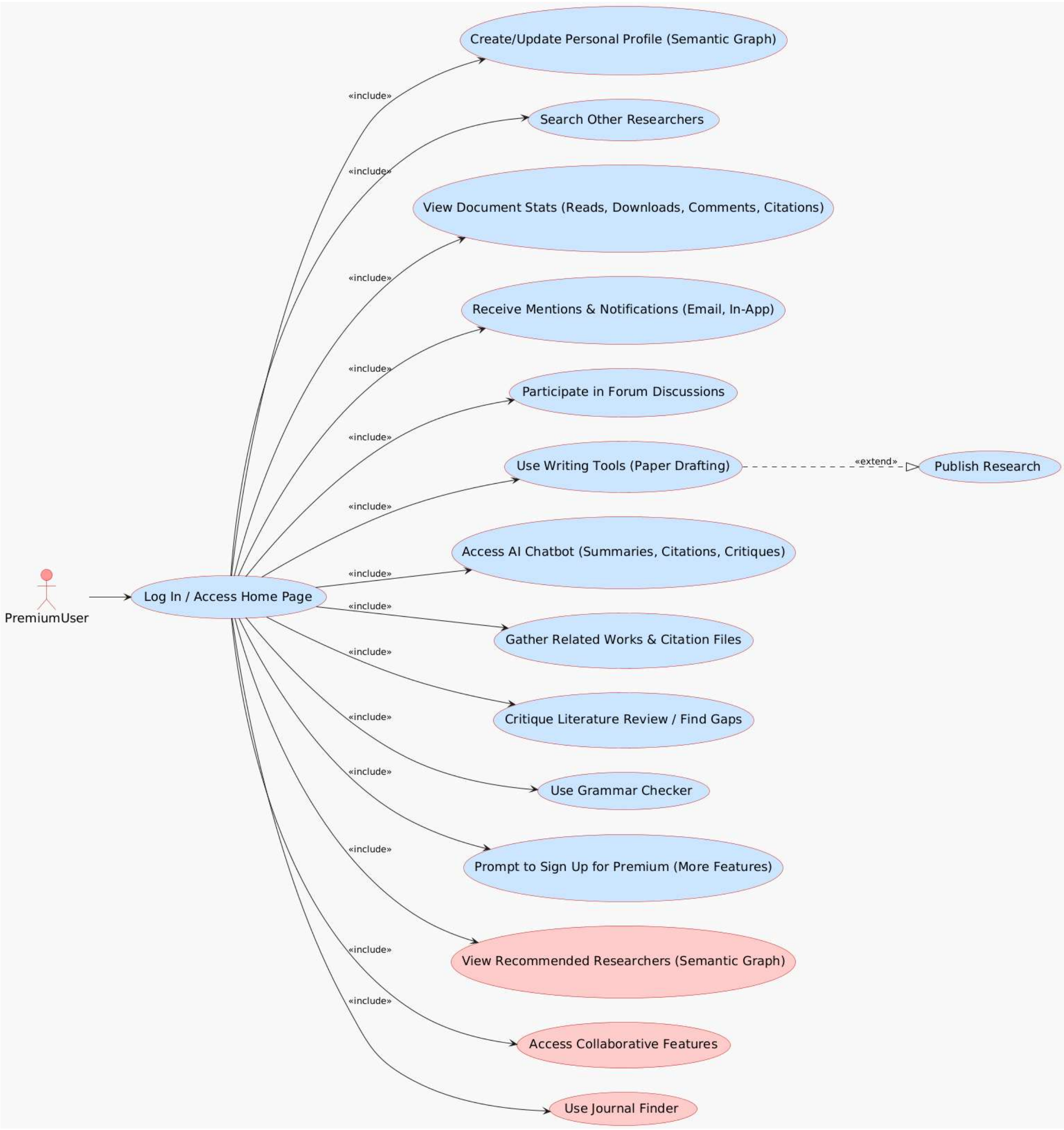


Figure 3.5 Premium user use case diagram

3.5.2 Use Case Description (Detailed Use Cases)

Use case description explains in detail the interaction between the user and the system, along with the steps or scenarios that may occur to the user while using the system.

- admin use case description

Advanced privileges login	
Actor	Admin
Description:	refers to the functionality where a user with administrative rights (admin) is granted access to the application with permissions that are higher or more extensive than those of regular users. These advanced privileges typically allow the admin to perform critical or sensitive actions that are restricted for standard users

admin-specific pages access	
Actor	Admin
Description:	refers to the capability of a user with administrative privileges to view and interact with web pages or sections of an application that are designed exclusively for managing and overseeing the system. These pages are typically restricted to authorized administrators and contain tools or features not available to regular users.

view list of all user accounts	
Actor	Admin
Description:	refers to the functionality that allows an administrator to access and manage a centralized view of all user accounts within the system. This feature provides an overview of the users categorized by their account types, enabling efficient management and oversight.

users violating rules suspend	
Actor	Admin
Description:	refers to the administrator's ability to enforce site policies by restricting access for users who breach the rules. This functionality allows the administrator to maintain the integrity of the platform and ensure a safe, respectful, and rule-compliant environment.

content control	
Actor	Admin
Description:	refers to the administrator's ability to manage and control the platform's content and functionality. This feature provides the flexibility to update, customize, or restrict site sections as needed, ensuring the platform stays relevant and well-maintained.

control feature access for all type of users	
Actor	Administrator user
Description	administrators can manage and restrict the access of various users to specific features

The ability to change site properties	
Actor	Administrator user

Description	The administrator has the ability to change site properties such as colour, logos, images, header and footer text.
Delete remove or pin anything	
Actor	Administrator user
Description	administrators can delete any offensive comment, inappropriate post removal or, pinning of the comment or post

Emergency notifications or announcements	
Actor	Administrator user
Description	The Administrator can push emergency notifications or announcements such as new feature release, planned downtime.

View system logs	
Actor	Administrator user
Description	The Administrator can view all system logs

- Registration user use case description

Log In	
Actor	**Registered User**
Description:	When the user creates an account can sign in and access the home page and then can access 95% of the features in the system.

Create/Update Personal Profile	
Actor	**Registered User**

Description:	The registered user can create personal profile and update it such as change password, username or even the email address he used to create the account.

Search Other Researchers	
Actor	**Registered User**
Description:	When the user is already has an account and sign in into it, he can search about other researchers that have an account too.

View Document Stats	
Actor	**Registered User**
Description:	The user can read the published researches download it, leave a comment, and citation of it.

Receive Mentions & Notifications	
Actor	Registered User
Description	The user is notified (via email and/or in-app alerts) when someone mentions them in a discussion or when system events relevant to their research occur (e.g., feedback on a paper, collaboration requests).

Participate in Forum Discussions	
Actor	Registered User
Description	The user can view, post, and respond to topics in the forum, engaging with other researchers through discussions, Q&A, or sharing insights.

Use Writing Tools (Paper Drafting)	
Actor	Registered User
Description	The user can create and edit research documents within the platform, utilizing features like formatting support, version control, and AI-driven writing aids.

Publish Research	
Actor	Registered User
Description	The user can finalize and submit a research document for publication on the platform, making it accessible to others (subject to any visibility settings or peer-review processes).

Access AI Chatbot (Summaries, Citations, Critiques)	
Actor	Registered User
Description	The user engages with the AI chatbot to retrieve literature summaries, generate citation references, or receive critique and suggestions on research-related queries.

Gather Related Works & Citation Files	
Actor	Registered User

Description	The user can compile and organize references pertinent to their research topic, exporting them as citation files (e.g., BibTeX) for easy integration into publications.

Critique Literature Review / Find Gaps	
Actor	Registered User
Description	The user utilizes AI or platform-supported analysis tools to evaluate their gathered literature, identify weaknesses or unexplored areas, and refine their review with stronger arguments or additional sources.

Grammar checker	
Actor	Registered user
Description	User use this tool for grammar detection and word correction

Sign up promotion	
Actor	Registered user
Description	The regular user prompted to sign up for premium advanced features.

- guest use case description

Read Research Papers	
Actor	Guest
Description:	The user can read the research published research papers without need to have an account in the system.

Watch Tutorial Video	
Actor	Guest
Description:	The user can watch a tutorial video about the system to know about it and learn how can use it.
Sign Up	
Actor	Guest
Description:	The user can sign up through filling their information (username, email address, password, etc.) so he can access all features in the system.

- Premium user use case description

View recommended researchers	
Actor	Premium user
Description	The user can be provided with recommended researchers working in same field or have the same interests through the similarity in the researcher profile graph.

Collaborative Features	
Actor	Premium user
Description	The user can collaborate with other researchers in the same field by using their graphs and use a collaborative space.

Journal Finder	
Actor	Premium user

Description	This feature provides the user with the appropriate publishing house or magazine for their research

3.5.3 Sequence Diagram

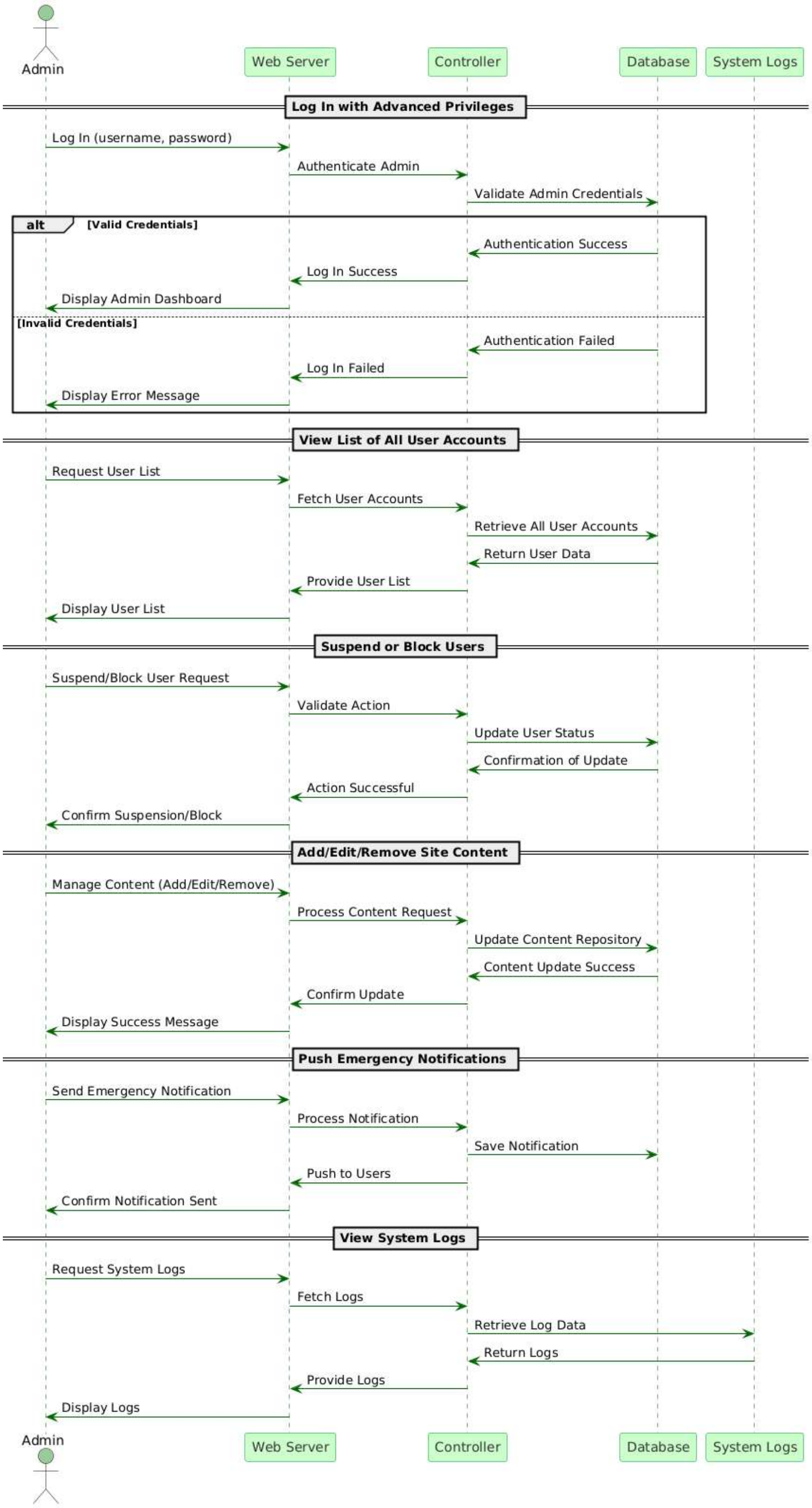
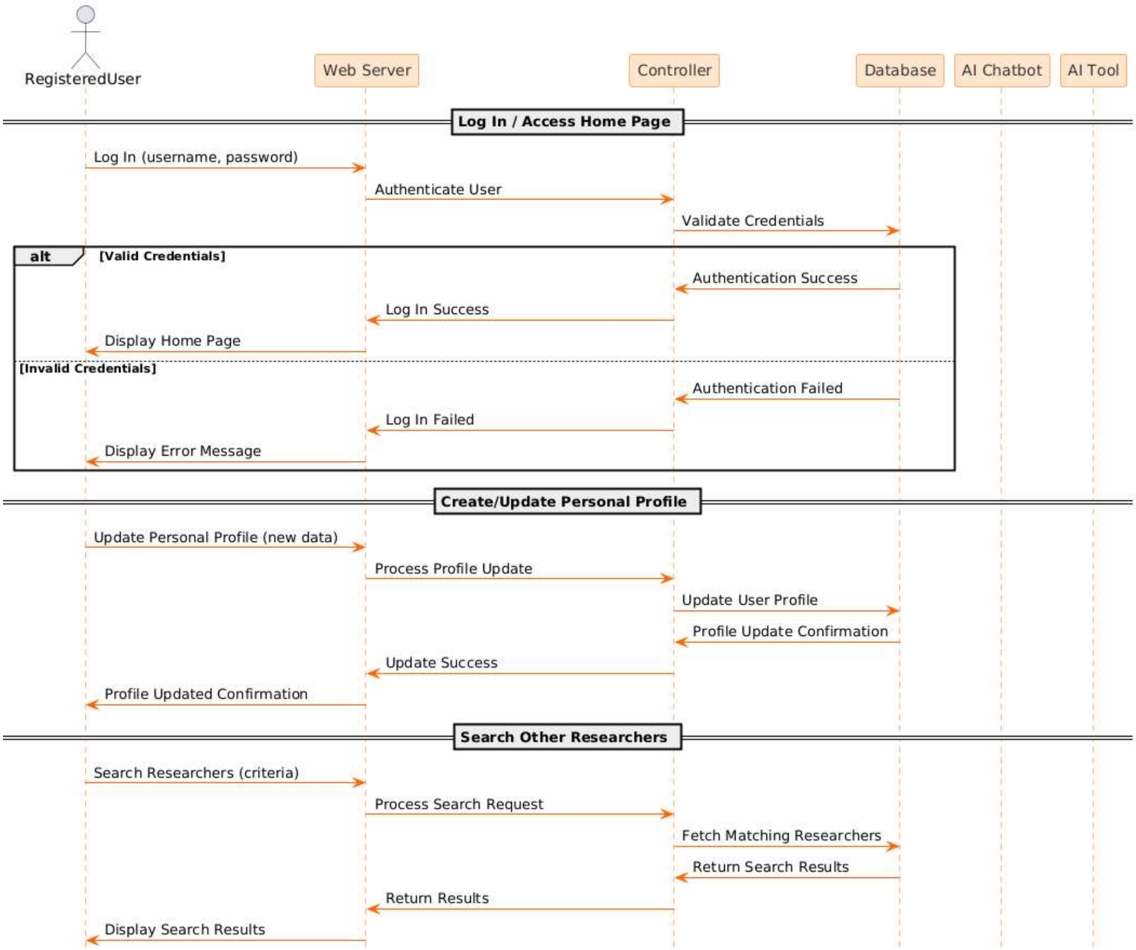
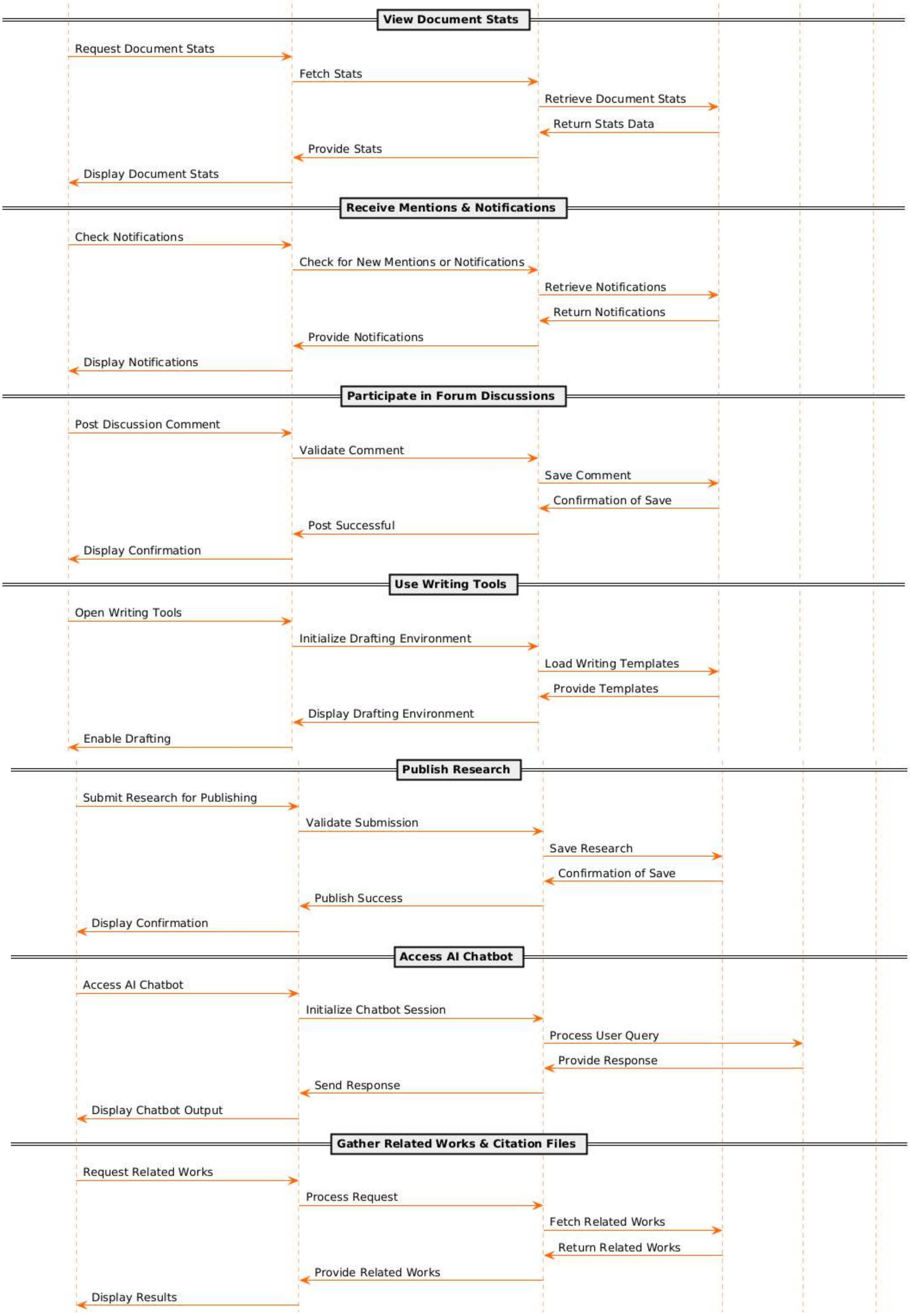


Figure 3.6 Admin Sequence Diagram


RegisteredUser
Web Server
Controller
Database
AI Chatbot
AI Tool
Log In / Access Home Page
Log In (username, password)
Authenticate User
Validate Credentials
alt
[Valid Credentials]
Authentication Success
Log In Success
Display Home Page
[Invalid Credentials]
Authentication Failed
Log In Failed
Display Error Message
Create/Update Personal Profile
Update Personal Profile (new data)
Process Profile Update
Update User Profile
Profile Update Confirmation
Update Success
Profile Updated Confirmation
Search Other Researchers
Search Researchers (criteria)
Process Search Request
Fetch Matching Researchers
Return Search Results
Return Results
Display Search Results


View Document Stats
Request Document Stats
Fetch Stats
Retrieve Document Stats
Return Stats Data
Provide Stats
Display Document Stats
Receive Mentions & Notifications
Check Notifications
Check for New Mentions or Notifications
Retrieve Notifications
Return Notifications
Provide Notifications
Display Notifications
Participate in Forum Discussions
Post Discussion Comment
Validate Comment
Save Comment
Confirmation of Save
Post Successful
Display Confirmation
Use Writing Tools
Open Writing Tools
Initialize Drafting Environment
Load Writing Templates
Provide Templates
Display Drafting Environment
Enable Drafting
Publish Research
Submit Research for Publishing
Validate Submission
Save Research
Confirmation of Save
Publish Success
Display Confirmation
Access AI Chatbot
Access AI Chatbot
Initialize Chatbot Session
Process User Query
Provide Response
Send Response
Display Chatbot Output
Gather Related Works & Citation Files
Request Related Works
Process Request
Fetch Related Works
Return Related Works
Provide Related Works
Display Results

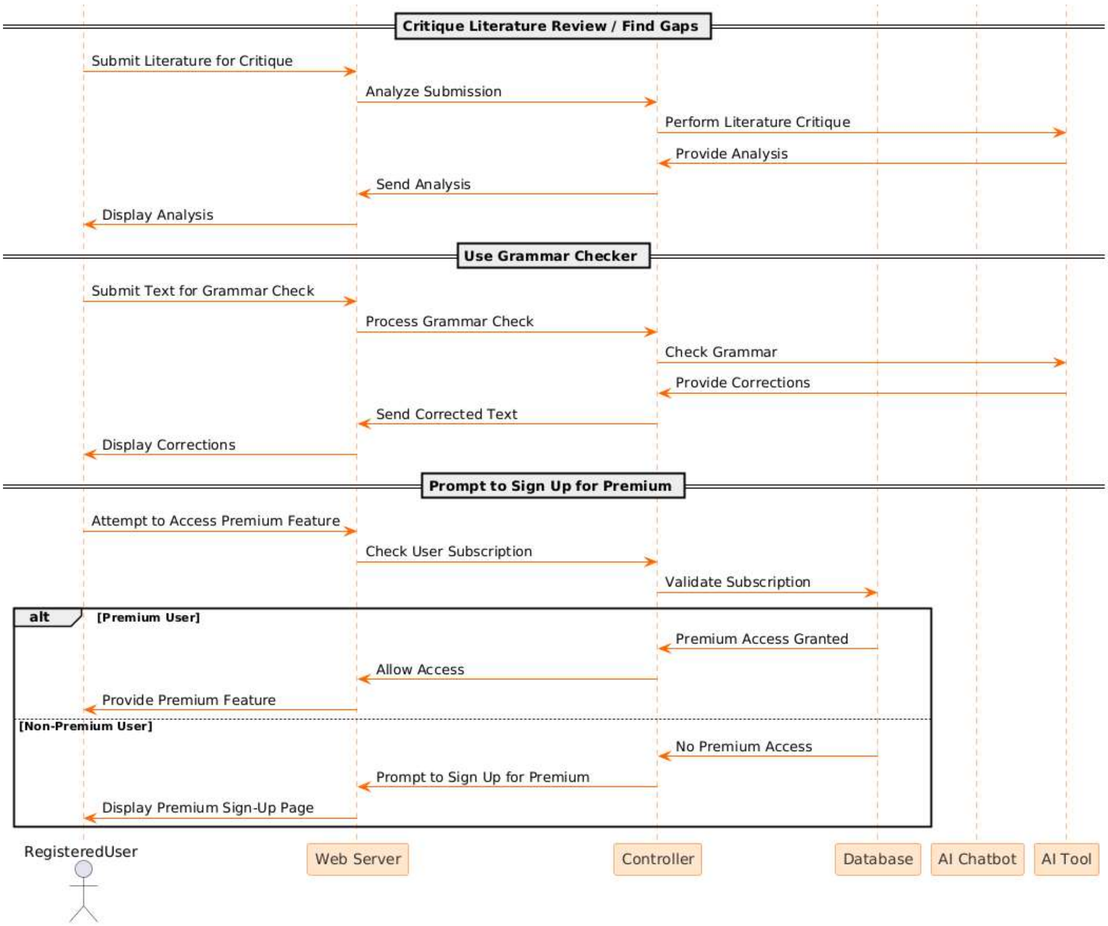


Figure 3.7 Registered user Sequence Diagram

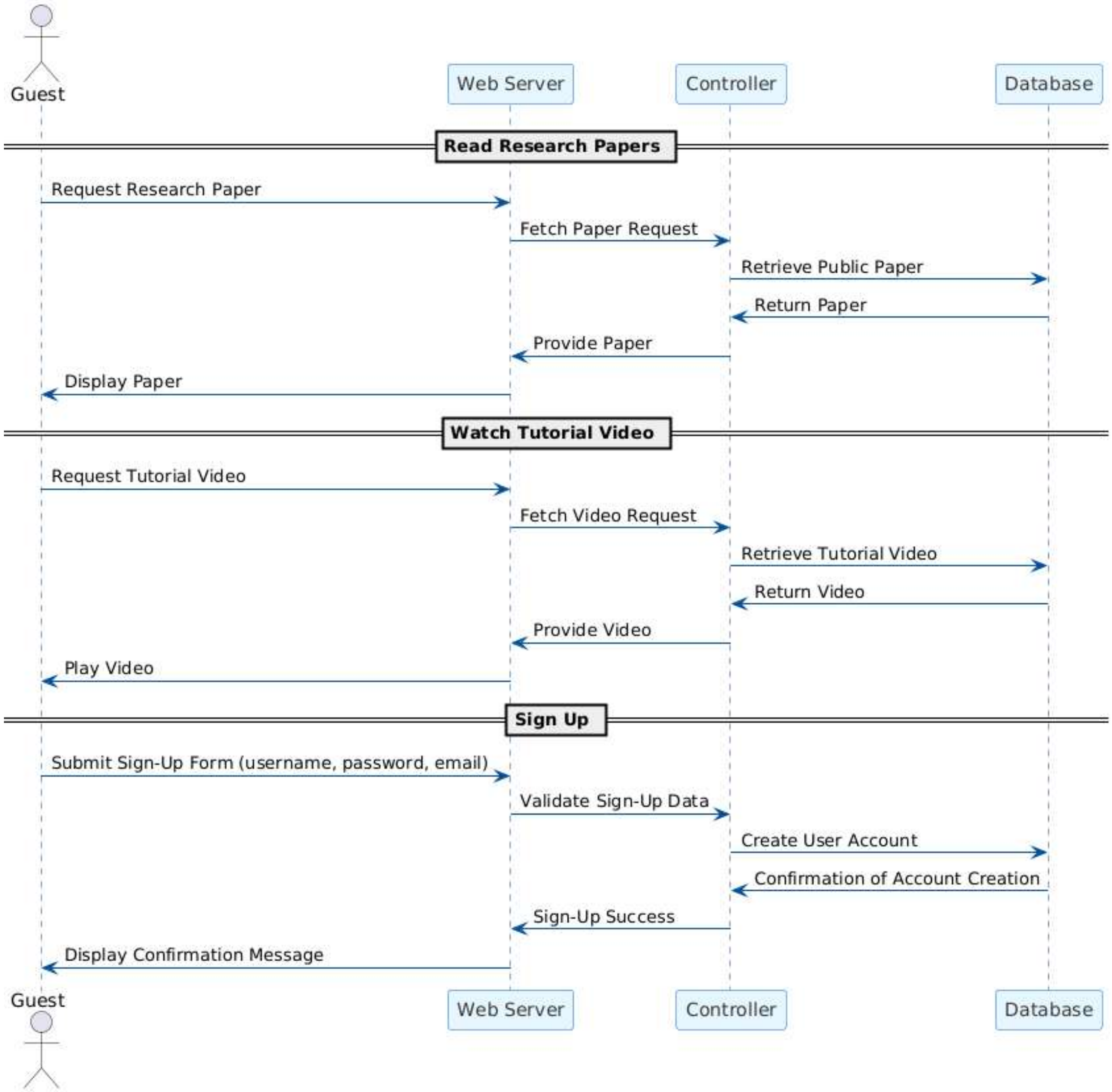


Figure 3.8 Guest Sequence Diagram

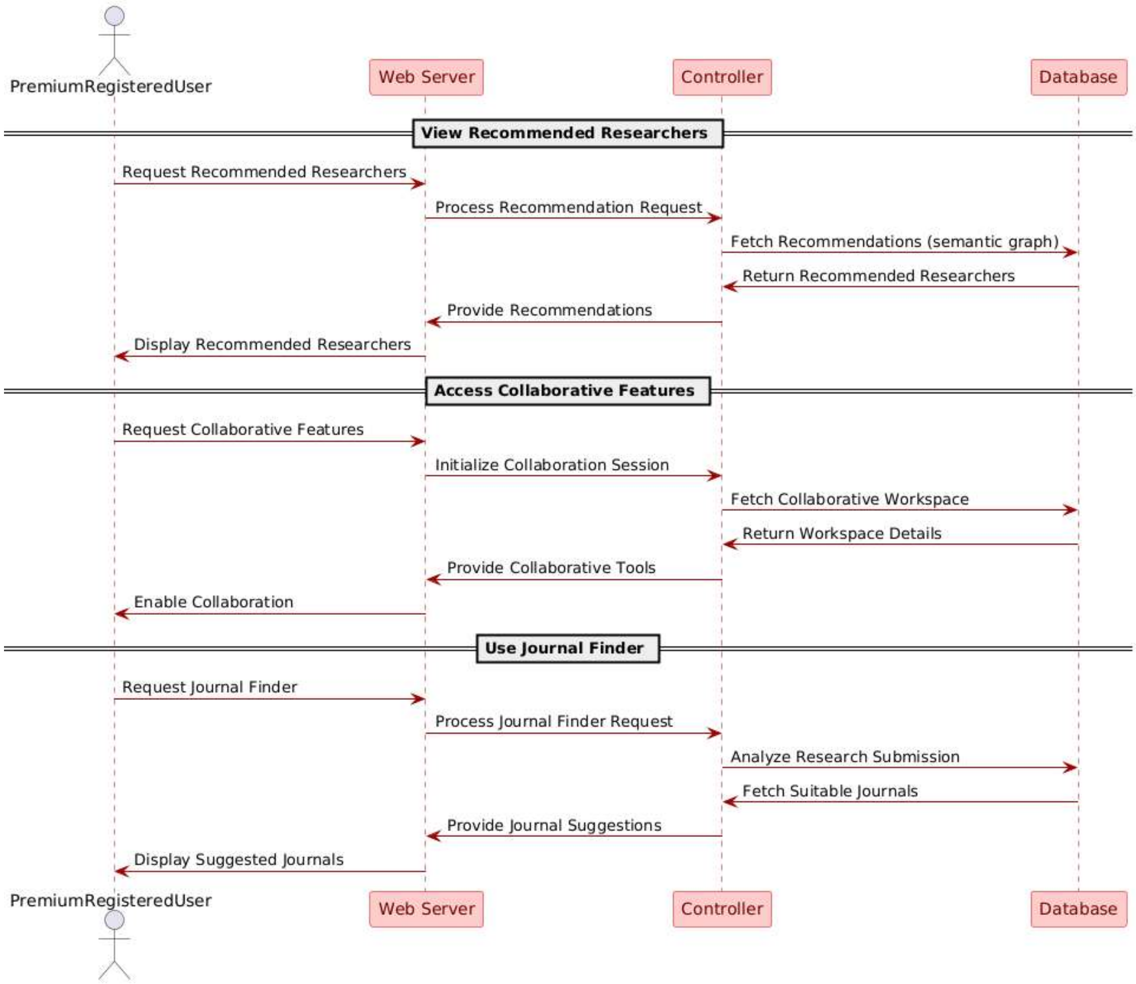


Figure 3.9 Premium user Sequence Diagram

3.6 Summary

This chapter explains how different people use and interact with the system and shows how processes work together and are arranged using use case diagrams, sequence diagrams, and illustrations.

Chapter Four: System Design

Each part of the challenge is formulated in a way that is illustrated in the design of the project. Every question or hypothesis must be researched and investigated, and if necessary, all variables must be identified and manipulated.

4.1 Introduction

In this chapter, we detail each component, modules and vital information of the system and ensure that they meet the criteria. Here we effectively represent the system features covered in other chapters, providing a useful description of what is expected to meet RESHUB's special requirements.

4.2 Class Diagram

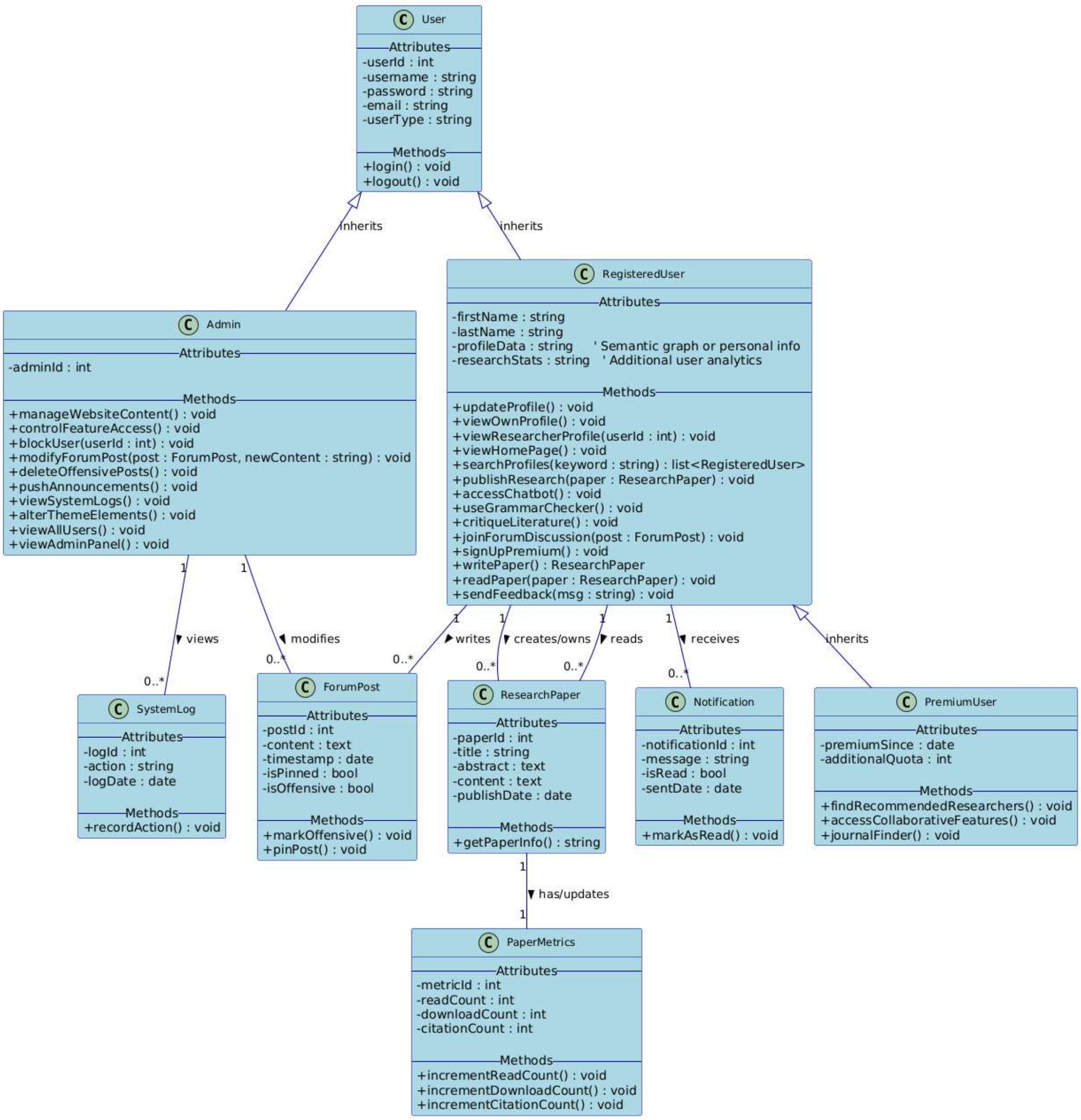


Figure 5.1 Class diagram

4.3 Database Design

4.3.1 Database Entity Relationship Diagram(ERD)

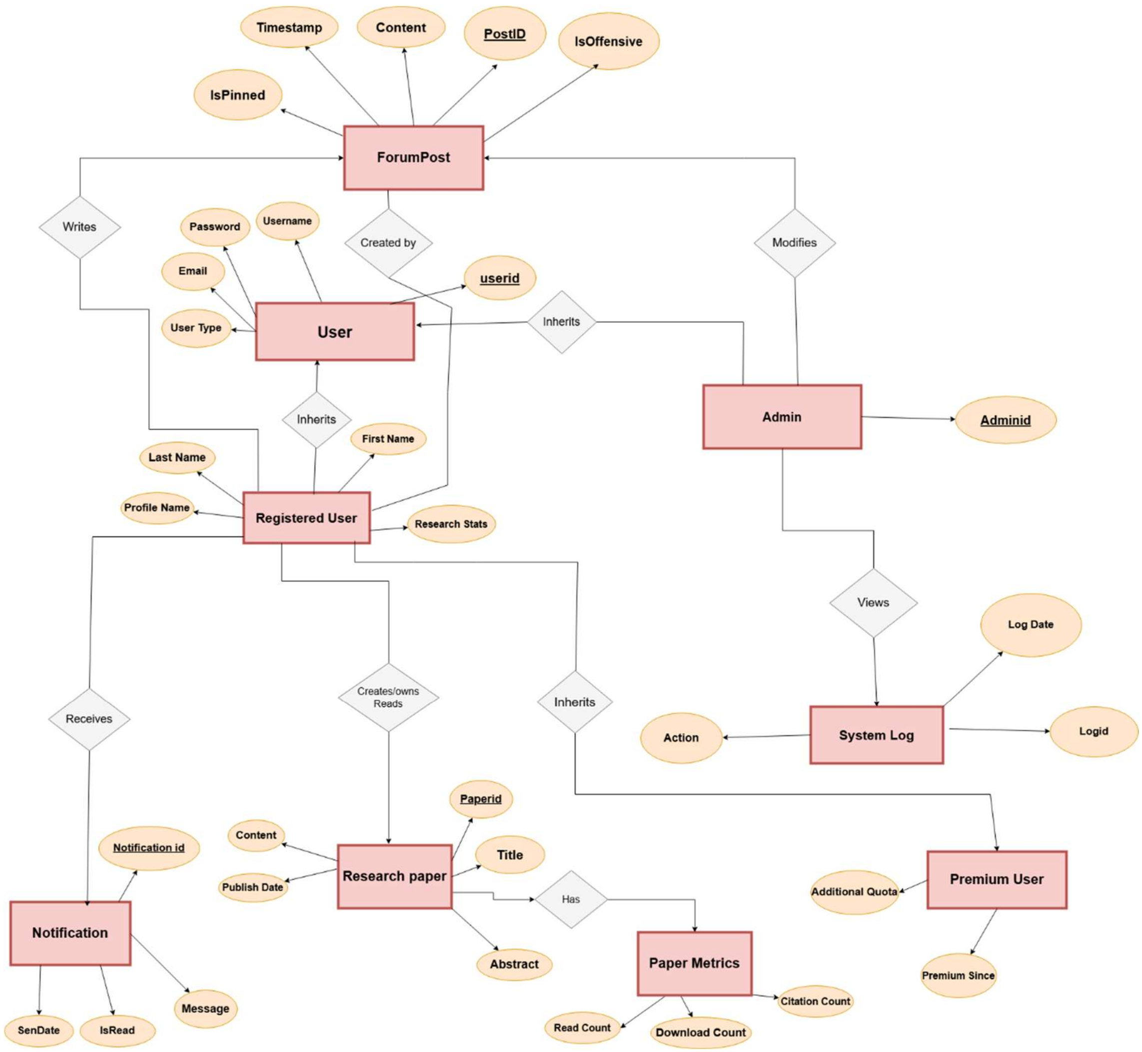


Figure 5.2 ERD diagram

4.3.2 Database Mapping

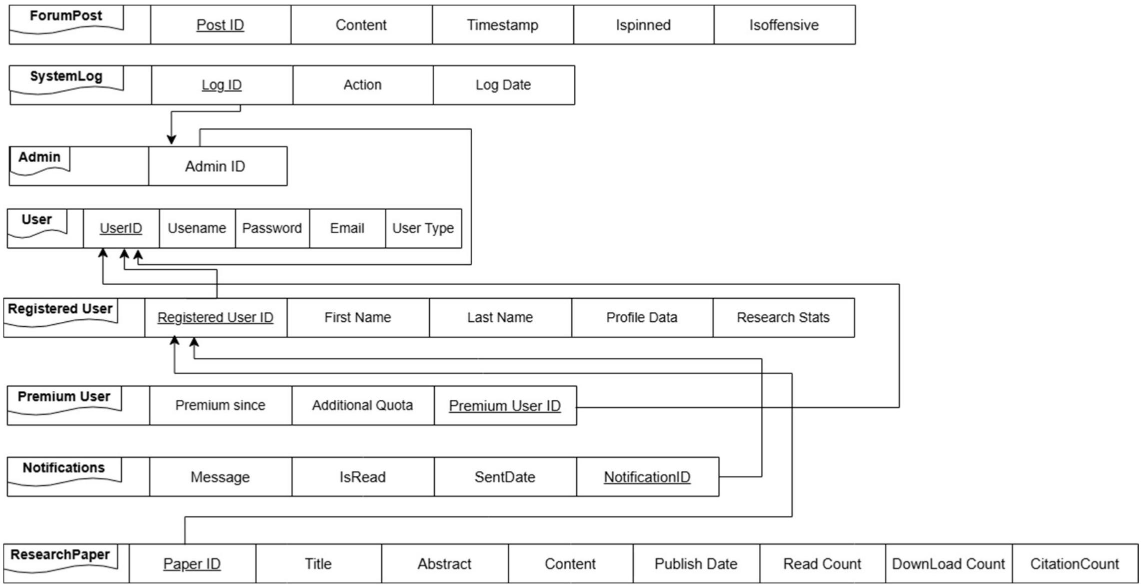


Figure 5.3 Database Mapping Design

4.3.3 Database Schema Design

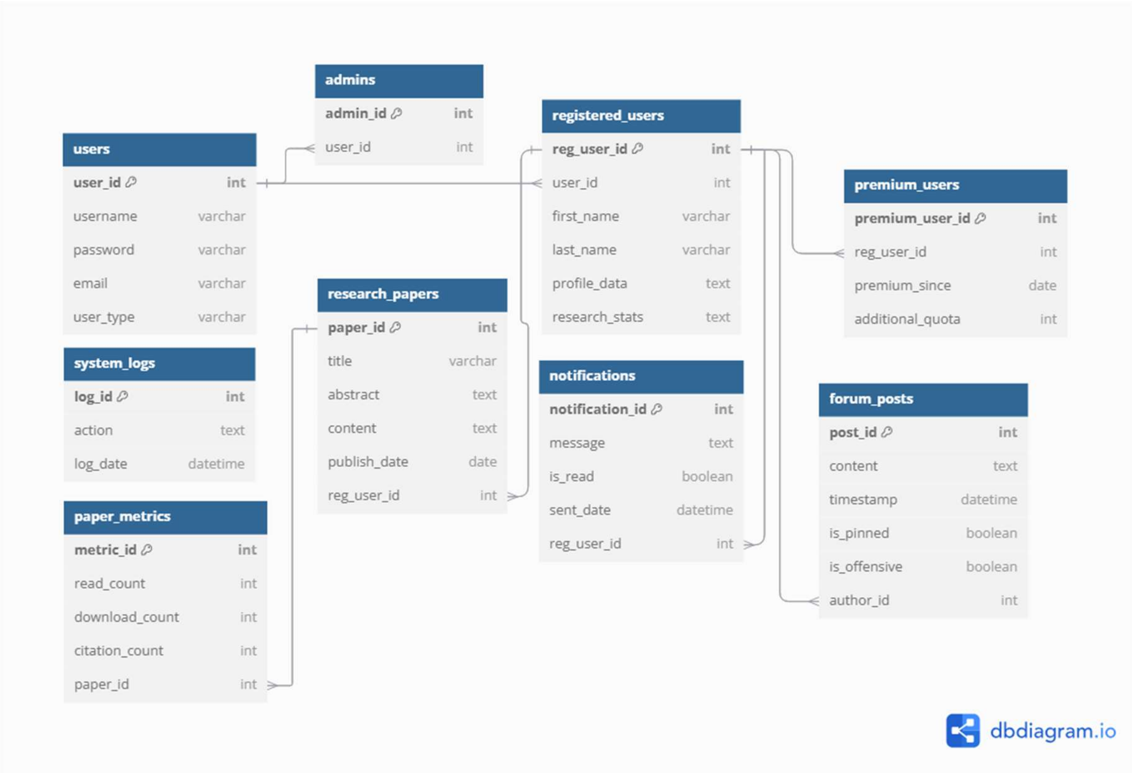
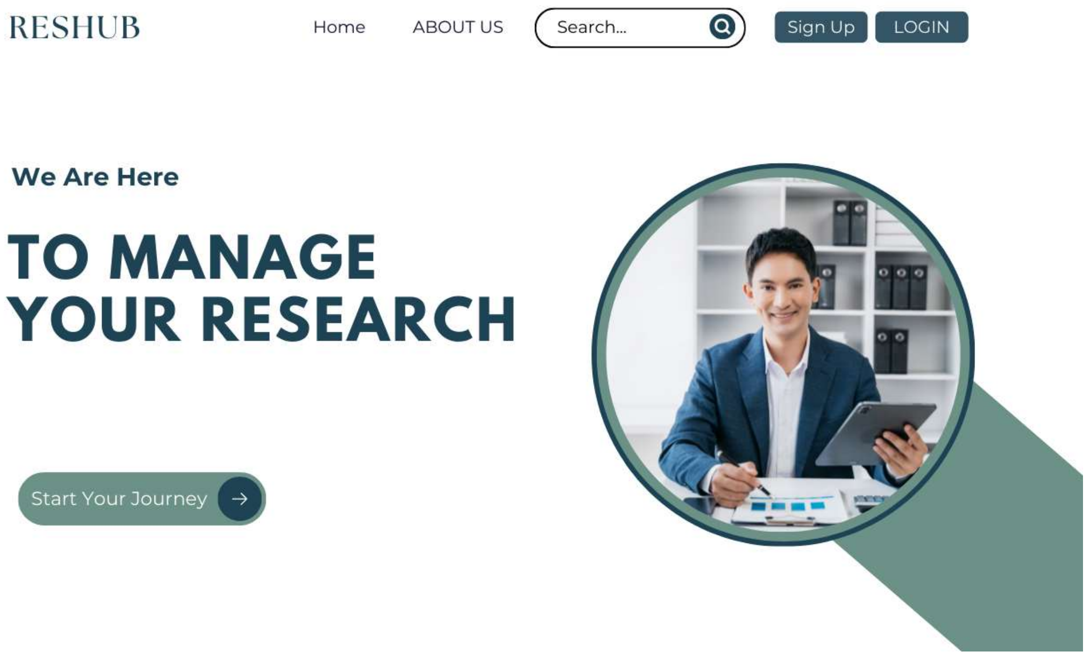


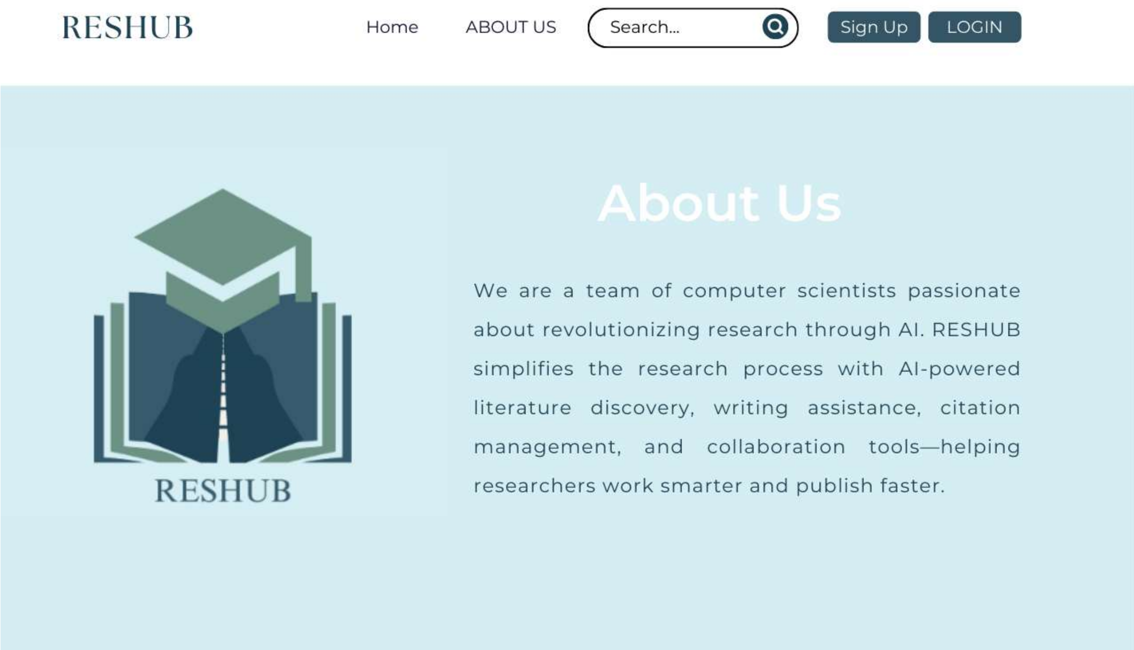
Figure 5.4 Database Schema Design

4.4 Interface Design

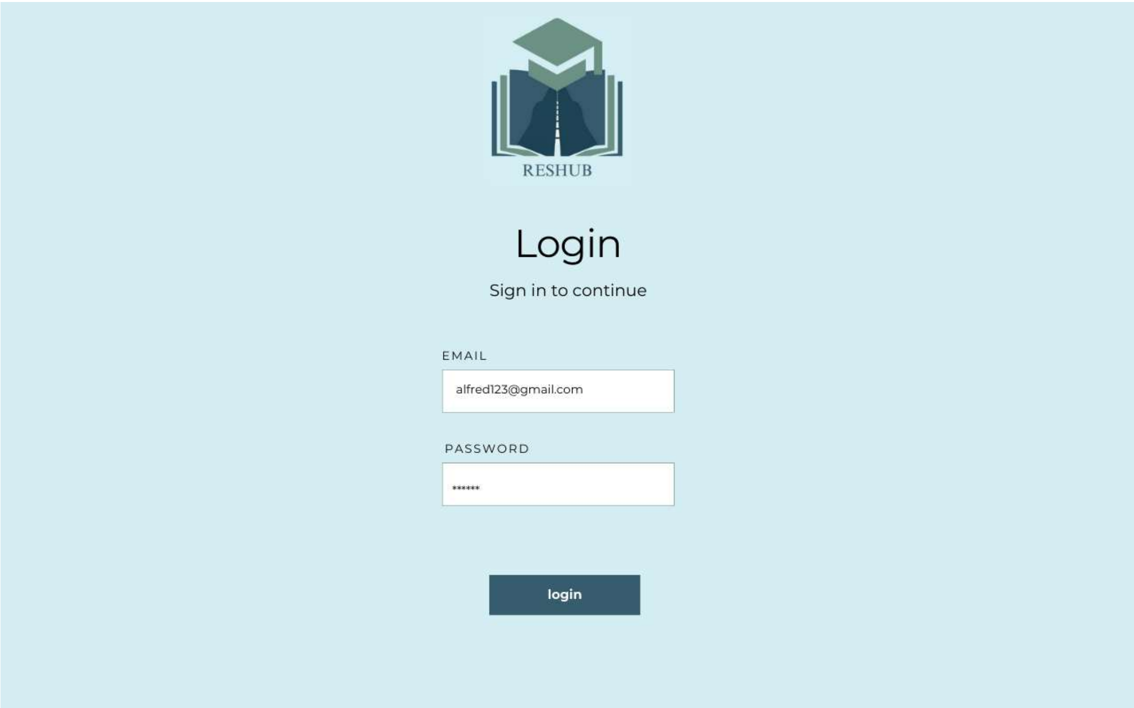
Home Page:


RESHUB
Home
ABOUT US
Search...
Sign Up
LOGIN
We Are Here
TO MANAGE
YOUR RESEARCH
Start Your Journey

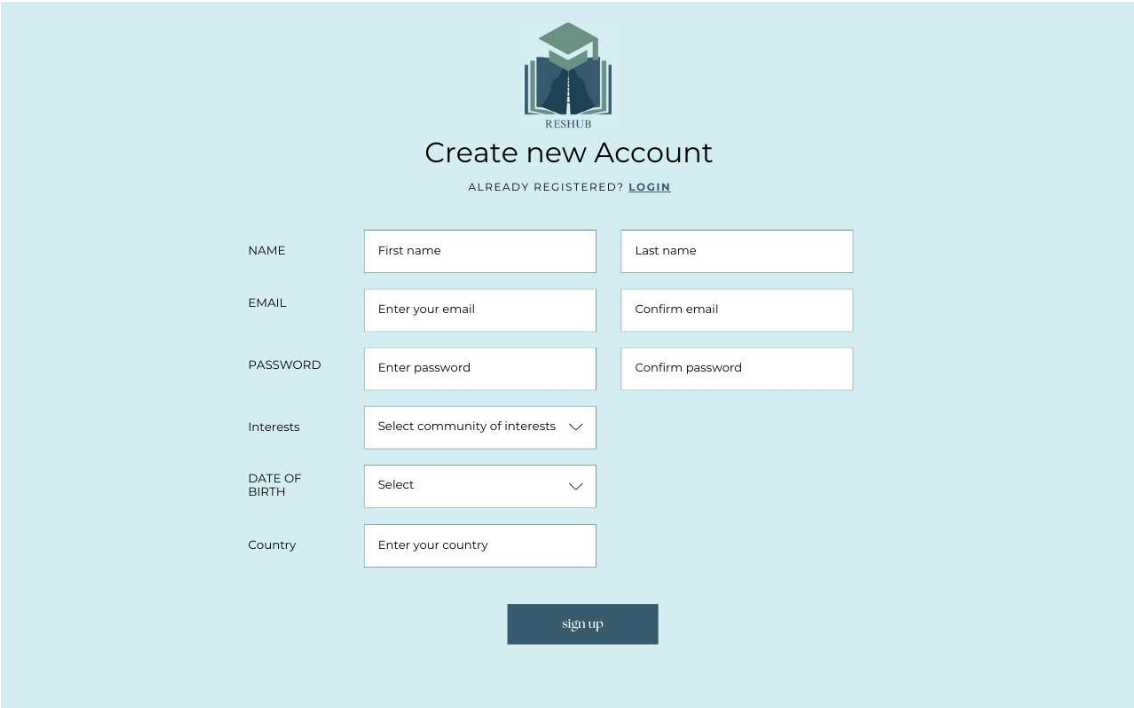
About Us:



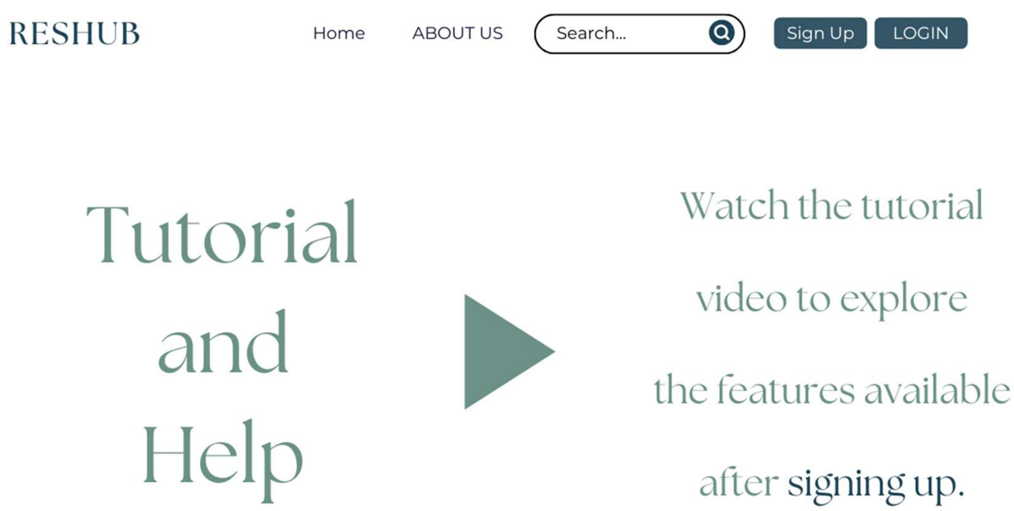
Login:



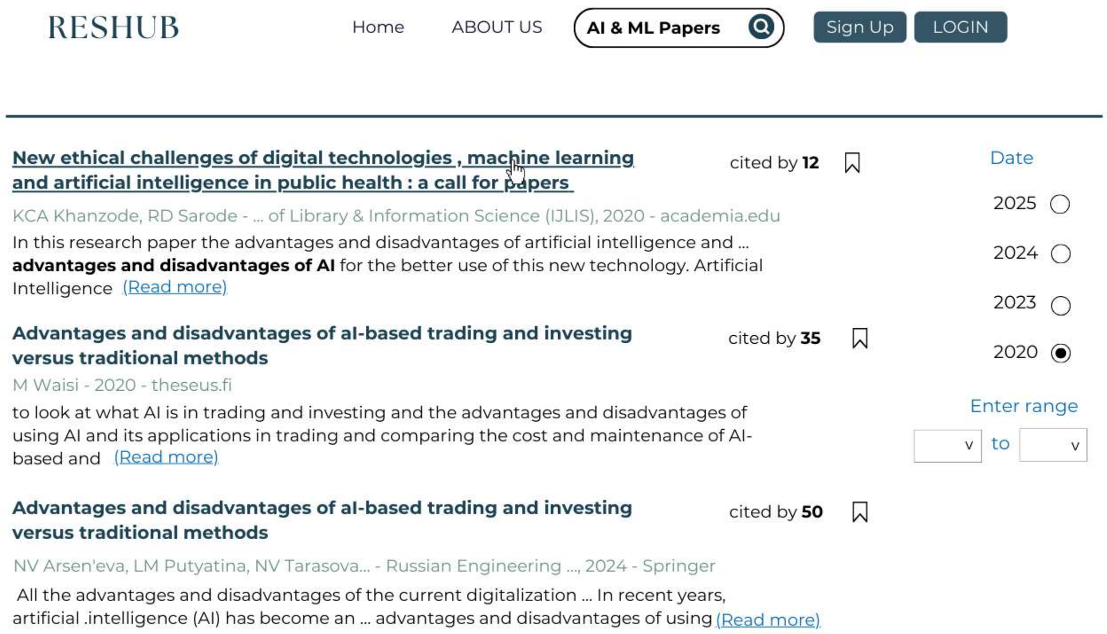
Registration:



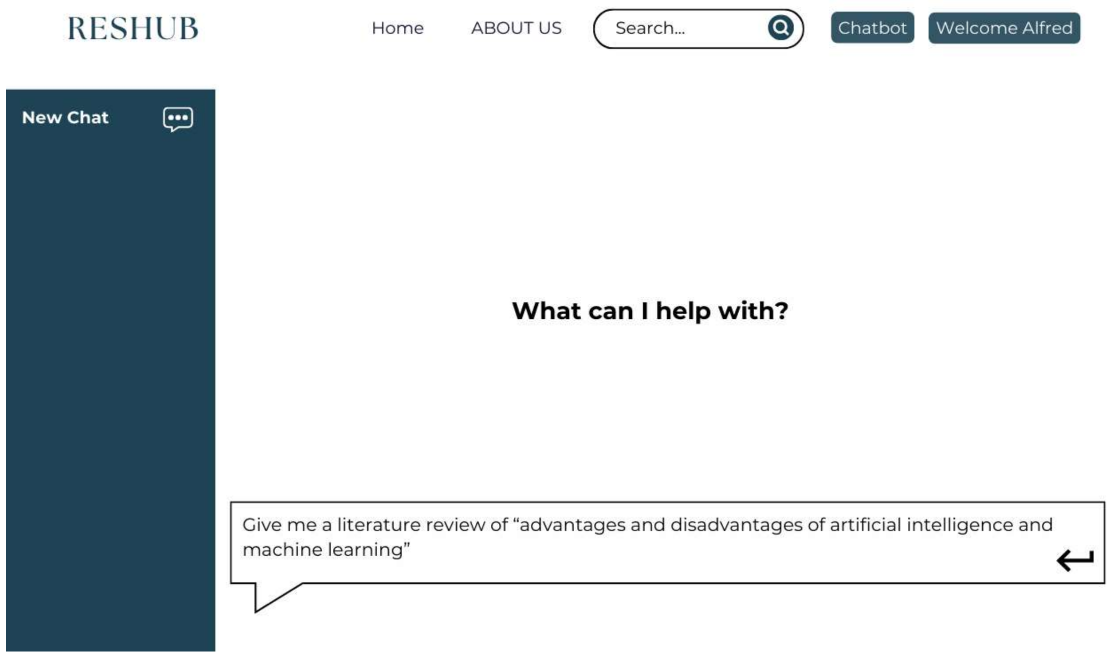
Tutorial:



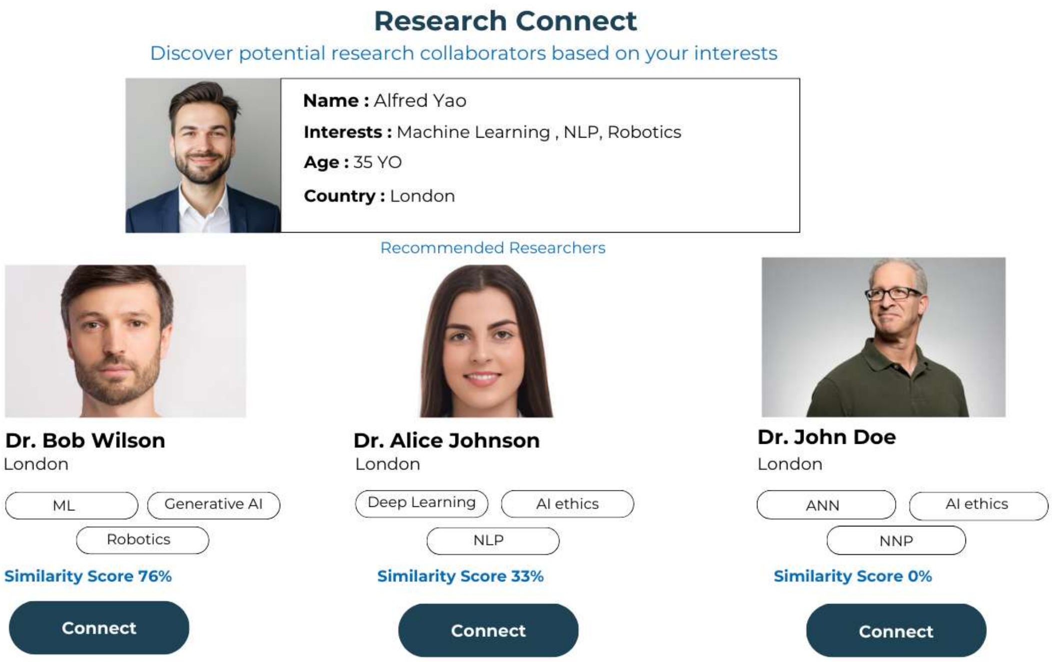
Search:



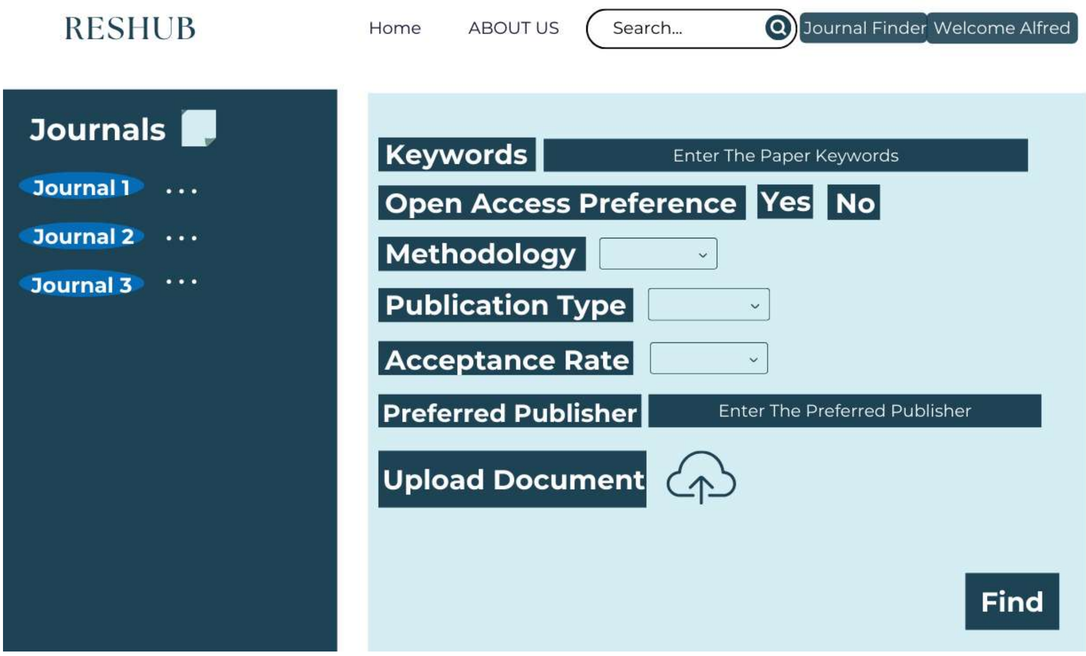
Literature Review Chatbot:



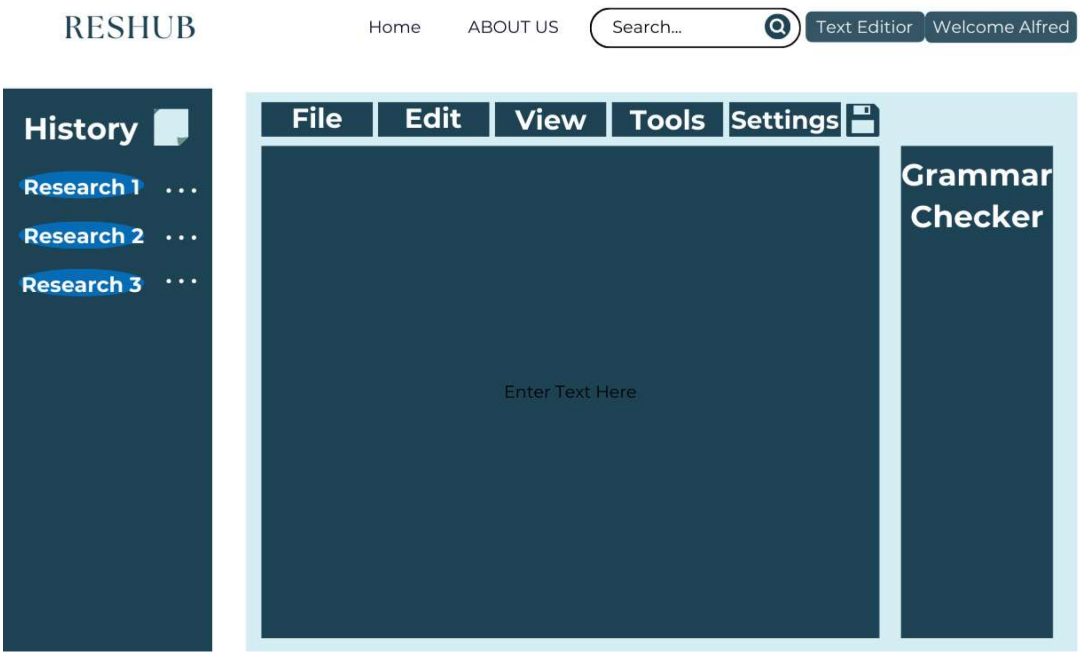
Research Recommendation:



Journal Finder:



Text Editor:



4.5 Summary

The database and designs are addressed in this chapter with the aim of identifying all the actual components of the system, creating user interfaces with elements that make them look more convincing and efficient to the user.

Chapter Five. Conclusions and Future Work

5.1 Conclusions

This project is a summary of what we learned during our studies at the College of Information Technology, which will be the first step towards integration into the labor market. Our project is based on greatly facilitating the researcher from various fields of practical research in most aspects of research, as it collects research related to their research, summarizes it, finds strengths and weaknesses, and compares it to their own research. Giving them some notes that their research should contain to distinguish it from other previous research and helps them write it correctly and with the correct grammatical rules. Thus, the researcher saves effort in collecting, understanding, comparing and improving research, working on preparing the practical research in an easy and smooth manner.

5.2 Future Work

On the current vision of the project, various paths are possible to be enhanced elevating from the experience of the platform, such as integrating more sophisticated NLP models or possibly fine tuning ready LLM models providing deeper more personalized results, expanding collaborative workspace features, adjusting user interface to be even more intuitive with support to multiple languages and the potential to collaborating with international institutions and publishers, ensuring a fully integrated research ecosystem.

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